Cost-Sensitive Attribute Reduction for Label Distribution Learning Using Granular-Ball Computing with Objectwise Reconstruction Bounds

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Abstract

Aggregate reconstruction criteria do not explicitly enforce fidelity for individual objects: a small mean error can coexist with increased reconstruction error for particular objects. This study formulates cost-sensitive attribute reduction for label distribution learning under fixed-reference objectwise bounds. Each training object's squared reconstruction error is constrained by its full-attribute reference error plus a prescribed tolerance. GBIP-O1 combines label-information geometry and granular-ball reconstruction with complete cost-ordered deletion sweeps. For a nondegenerate reference, the algorithm preserves feasibility and terminates at an empty subset or a feasible subset that is irreducible under single-attribute deletion. A label-margin condition provides a sufficient guarantee for preserving a unique dominant label. Evaluation on 25 tasks spanning 16 feature families and a separate SCUT-FBP5500 image pipeline examines simulated acquisition budgets, reconstruction error, prediction and resource use. The results show that mean preservation can permit individual deterioration, whereas strict objectwise preservation can lead the deletion search to retain all attributes. Matched-constraint comparisons and sensitivity analyses characterize the

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effects of search strategy, geometry, sample size, prices and tolerance. The resulting guarantees apply to the specified training reconstruction constraints; held-out prediction and end-to-end resource use require separate evaluation.

Keywords: Label distribution learning, cost-sensitive attribute reduction, objectwise reconstruction, granular-ball computing, directed dissimilarity

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1. Introduction

Label distribution learning (LDL) represents an instance by a distribution over labels, retaining their relative description degrees rather than assigning a single label [1]. Reducing attributes can lower acquisition or storage requirements [2, 3], but can change a reconstructed label distribution. An aggregate reconstruction criterion allows improvements for some objects to compensate for deterioration for others. We therefore seek lower-cost attribute subsets subject to an upper bound on each training object's squared reconstruction error relative to its fixed full-attribute reference error. This condition controls error against the observed labels; it does not directly bound the distance between the two reconstructed distributions. Our main benchmark uses simulated attribute prices, while the image case measures stored-feature bytes and processing time separately.

Facial attractiveness assessment illustrates the distinction between aggregate accuracy and individual reconstruction error. Rating distributions describe variation among observers that a single mean score can obscure [4, 5]. Reducing stored facial descriptors can increase the reconstruction error for a particular image even when the dataset mean error decreases. A protocol that limits each stored instance's error must therefore check the corresponding instance-level inequalities. Such a requirement is distinct from preserving each reconstructed distribution or its dominant rating category.

Record-specific reconstruction-error limits may be relevant to medical imaging, credit assessment or industrial inspection if an application defines a reference, loss and tolerance. Work on hidden stratification shows that aggregate metrics can conceal subgroup failures [6]; it does not establish the utility of the particular objectwise constraint studied here. Model cards motivate transparent reporting [7]. In our setting, reporting the reference error, allowed increase and measured error for each inspected record makes the

specified constraint auditable. Its practical value still requires application-specific validation of those choices.

The relevance of an objectwise reconstruction bound is further contingent upon the margin of separation between label distribution degrees. When the two largest label degrees are close in magnitude, even a minor perturbation can alter the dominant label, implying that a reconstruction-error bound does not inherently guarantee label or ranking stability [8]. To bridge this gap, we relate the bound directly to the object's label margin, establishing a formal sufficient condition to preserve its unique dominant label. This theoretical guarantee is complemented by empirical assessments of dominant-label, top-two, and within-object ranking shifts, distinguishing the properties certified by the reconstruction constraint from those necessitating independent evaluation.

Existing LDL feature-selection methods predominantly evaluate attribute subsets through dependency, local consistency, or conditional discrimination. Section 2 reviews these approaches alongside information-based selection, granular-ball representations, and acquisition-cost models. GBFRS combines granular-ball fuzzy dependency with local consistency [9], while IWDLFS and IWCLFS employ dependence and conditional discrimination criteria within an informational weighted neighborhood rough-set model [10]. These formulations assess the discriminative structure retained by an attribute subset. In contrast, our formulation addresses a different requirement: whether every training instance's reconstruction error remains bounded by its full-attribute reference error plus a prescribed tolerance. These objectwise bounds define the feasible family, within which GBIP-O1 seeks lower-cost subsets through complete deletion sweeps. Its termination certificate establishes single-deletion irreducibility rather than global cost optimality.

To implement this formulation, we integrate label-information weights and directed dissimilarity into granular-ball reconstruction. Beginning from the full attribute set, the GBIP-O1 algorithm systematically evaluates attribute deletions in descending order of cost, accepting a deletion only when all objectwise bounds remain satisfied. Because removing attributes regenerates the granular balls, distance monotonicity does not guarantee the monotonicity of reconstruction feasibility. Consequently, rejected deletions are revisited across successive complete sweeps until an entire sweep yields no accepted deletions. For a nondegenerate reference, the algorithm terminates at Empty or a feasible nonempty subset that is single-deletion irreducible.

This study contributes to the literature in three respects.

1. A cost-sensitive LDL attribute-reduction framework is formulated under fixed-reference objectwise reconstruction bounds, distinguishing individual feasibility from aggregate quality and defining acquisition-cost minimization over the feasible family.

2. GBIP-O1 integrates label-information geometry, granular-ball reconstruction and complete cost-ordered deletion sweeps. The analysis establishes feasibility preservation, finite termination and single-deletion irreducibility, together with a sufficient label-margin condition for preserving a unique dominant label.

3. Evaluation on 25 tasks spanning 16 feature families and a separate image pipeline characterizes cost–preservation trade-offs through matched-constraint comparisons, geometry ablations and sensitivity analyses. Decision diagnostics and resource measurements distinguish reconstruction guarantees from predictive and computational outcomes.

Study Overview. Figure 1 links the reconstruction objective, training protocol, subset search and empirical assessment. Subsets and predictors are fitted on the designated training partitions and frozen before their recorded held-out evaluation; comparator parameters use training-internal cross-validation (Section 7). Geometry fitting and deletion acceptance do not read test labels. Because the datasets and development results had previously been examined, this separation within each run does not make the study independent confirmation. Training reconstruction, held-out prediction and image-pipeline resources are distinct outcomes. Figure 3 details the deletion control flow.

The comparator branch includes GBFRS-EQ, IWDLFS, IWCLFS and GBFRS+O1. Mean uses average training-error preservation; Full and Empty use all attributes and no attributes, respectively. Evaluation covers theoretical guarantees, training feasibility, acquisition cost, label decisions, held-out Brier error, sensitivity, descriptor storage and measured query time. The prediction transformations are fitted on training data before being frozen.

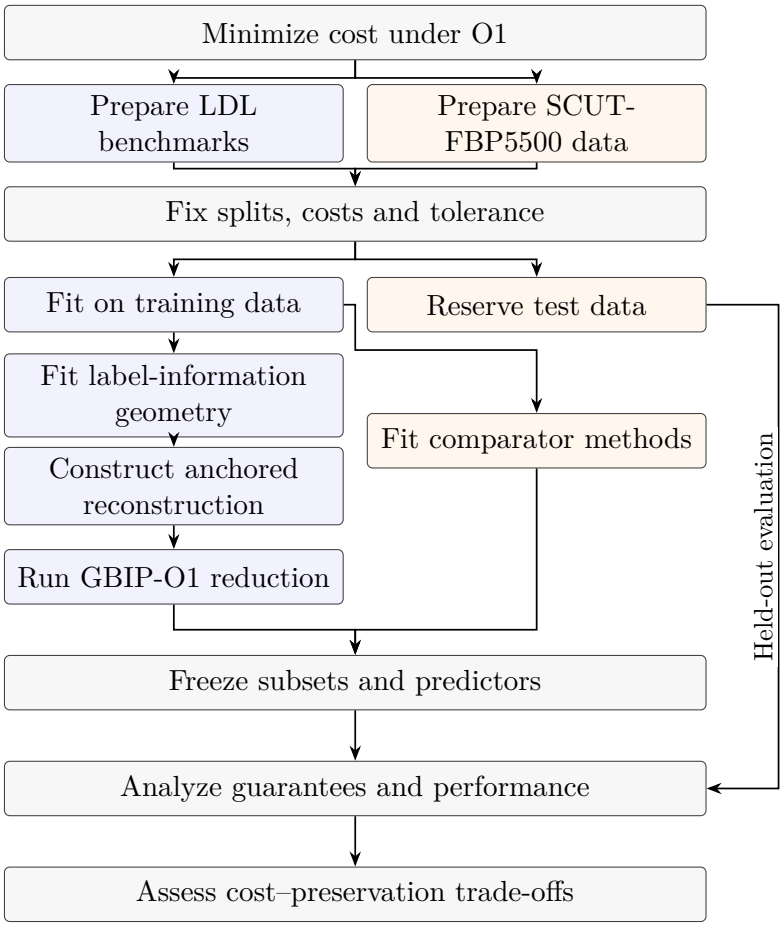


Figure 1: Research workflow within each recorded run: selectors and transformations are fitted on training data, and test data are used at the scoring stage. Development use of the datasets and splits is disclosed in the experimental setup. The image and array protocols are separate.

2. Related Work

2.1. Feature Relevance and Distribution-Valued Supervision

Feature selection distinguishes feature ranking, subset evaluation and the search used to choose a subset [11]. Minimum-redundancy maximum-relevance selection balances relevance against inter-feature dependence [12], while conditional-likelihood analysis relates information-based heuristics to a common statistical objective [13]. A relevance score, predictive loss and per-object reconstruction constraint measure different properties; agreement on one need not imply agreement on the others.

LDL supplies relative description degrees rather than a binary set of relevant labels [1]. Label enhancement addresses the related task of recovering distributions from logical labels using feature structure and label correlations [14]. Our setting instead treats the supplied training distributions as fixed. It does not infer missing degrees or validate whether enhanced labels provide an appropriate application reference.

2.2. Feature Selection for Label Distribution Learning

Several lines of research have adapted feature evaluation frameworks to distribution-valued labels. For instance, Qian et al. incorporate feature similarity and label correlation into LDL feature selection [15], Xiong et al. combine label distributions with fuzzy mutual information [16], and Deng et al. evaluate features using dual-similarity neighborhood fuzzy entropy [17]. More recently, You et al. construct fuzzy similarity via the statistical distribution of feature values to evaluate subsets using fuzzy mutual information [18]. These formulations use distribution-valued supervision in their feature criteria; they do not impose the fixed-reference objectwise reconstruction inequalities studied here.

Neighborhood and fuzzy rough-set approaches provide other feature criteria. Wu et al. use neighborhood relations in label-distribution space and label-weighted dependency [19]; Deng et al. incorporate label-significance distributions into neighborhood fuzzy rough sets [20]; and Lu et al. combine hierarchical structure with neighborhood granularity [21]. Lin et al. learn feature weights through an LDL prediction model with label-correlation regularization [22], while Shu et al. consider label-specific features [23]. These methods use different representation and evaluation criteria and are not interchangeable implementations of one objective.

Incomplete labels and streaming features define additional settings. Qian et al. use a neighborhood-tolerance discrimination index for incomplete distributions [24]; Chen et al. study online selection with label importance and correlation [25]; and Qian et al. combine granular-ball fuzzy rough sets with semi-supervised selection under incomplete labels [26]. Our training collection and available attributes are fixed before reduction. Missing supervision or incoming features would require a separate rule for constructing or updating the full-attribute reference.

The closest experimental comparators in this study are GBFRS, which integrates granular-ball fuzzy dependency with local consistency [9], and IWDLFS/IWCLFS, which employ dependence and conditional discrimination within an informational weighted neighborhood rough-set model [10]. We distinguish the original selection criteria from GBFRS+O1, the O1-filtered prefix comparator based on the GBFRS-EQ ranking. Meanwhile, the remaining studies reviewed above serve to establish broader methodological context rather than acting as additional empirical baselines. The primary contribution of GBIP-O1 lies in its explicit fixed-reference feasibility requirement and certified deletion endpoint, rather than a generalized claim of predictive superiority over all existing LDL feature-selection paradigms.

2.3. Granular-Ball Representation and Rough-Set Reduction

Rough sets characterize the information retained by an attribute subset through approximation and discernibility [27], with neighborhood formulations extending this perspective to heterogeneous feature spaces [28]. Granular-ball learning complements this by replacing uniform object-level representations with adaptive groupings. For instance, Xia et al. develop efficient and adaptive granular-ball generation methods [29], GBRS connects granular-ball learning with both Pawlak and neighborhood rough-set models [30], and GBG++ further investigates the computational efficiency and stability of ball generation [31]. These studies develop granular representation and construction mechanisms. Their classification or rough-set criteria do not by themselves certify the reconstruction constraints defined here.

In LDL, regenerated granular balls jointly influence the neighborhood structure and label-distribution reconstruction. Consequently, the mere geometric fact that removing nonnegative distance terms cannot increase the corresponding directed dissimilarity does not guarantee the monotonicity of the reconstruction criterion following regrouping. This fundamental limitation motivates the explicit feasibility evaluations and iterative complete

sweeps implemented in GBIP-O1. The analysis establishes that its termination certificate strictly pertains to the local irreducibility of the final subset under single-attribute deletions; it does not establish global cost minimization, the uniqueness of a reduct, or invariant endpoints across alternative ball-generation procedures.

2.4. Acquisition Cost and the Preservation Requirement

Feature acquisition cost differs from the loss caused by a prediction error [2]. Test-cost-sensitive reduction assigns attribute acquisition costs [32], and test-cost-constrained selection imposes a resource budget [3]. Decision-theoretic rough-set reduction also incorporates decision costs [33], while cost-sensitive random-forest selection accounts for feature measurement costs [34]. These distinctions require specifying both the minimized resource and the property to be preserved.

In our formulation, the objective function corresponds to the sum of selected attribute acquisition costs, whereas feasibility is governed by a distinct reconstruction-error inequality for each training instance relative to a fixed full-attribute reference. Consequently, aggregate relevance or error criteria cannot serve as substitutes for verifying these individual inequalities. Conversely, satisfying these objectwise bounds does not inherently guarantee lower held-out prediction loss or reduced total system latency. To account for this, the image-pipeline experiments report descriptor storage and end-to-end processing timing as separate performance metrics. Furthermore, our framework deliberately defers the scientific choice of the reference and tolerance parameters to the specific application protocol, with our experiments investigating the downstream consequences of these parameter choices rather than advocating for a universal acceptance threshold.

3. Preliminaries

3.1. Test-Cost-Sensitive Label Distribution Information System

Let $O = \{o_1, \ldots, o_n\}$ be a finite, non-empty set of objects, $A = \{a_1, \ldots, a_m\}$ a finite, non-empty set of conditional attributes, and $\mathcal{Y} = \{y_1, \ldots, y_q\}$ a finite, non-empty set of labels. Each attribute $a \in A$ induces a mapping $a : O \to V_a$, where V_a denotes the domain of attribute a. To handle hybrid feature spaces, we partition A as

$$A = A_n \cup A_c, \qquad A_n \cap A_c = \varnothing,$$

where A_n and A_c designate numerical and categorical attribute subsets, respectively. Either subset may be empty.

Definition 1 (Label Distribution Information System)**.** *A label distribution information system is a tuple*

$$\mathcal{S}_L = (O, A, \mathcal{Y}, L),$$

where $L = (\ell_i^k)_{n\times q}$ is the label distribution matrix satisfying the simplex constraints:

$$\ell_i^k \in [0,1], \qquad \sum_{k=1}^{q} \ell_i^k = 1 \quad (i = 1, \ldots, n;\ k = 1, \ldots, q). \tag{3.1}$$

The label distribution associated with object o_i is represented by the vector

$$\boldsymbol{\ell}(o_i) = (\ell_i^1, \ldots, \ell_i^q), \tag{3.2}$$

where ℓ_i^k quantifies the description degree of label y_k for object o_i [1].

The complete vector $\boldsymbol{\ell}(o_i)$ serves as the supervisory signal. The formulation uses the description degrees directly rather than replacing them by a dominant-label decision. Its reconstruction bound limits squared error in that vector; it does not guarantee unchanged label degrees or rankings. Description degrees need not have the interpretation of posterior class probabilities.

The proposed information system builds upon Pawlak's classical rough set model [27], incorporating neighborhood-based extensions specifically designed for hybrid data types [28, 35, 36]. While traditional attribute reduction techniques rely on heuristic searches guided by dependency or information measures over crisp equivalence classes [37, 38, 39, 40], the present formulation differs from those crisp-system settings in two respects: its decision representation is distribution-valued, and its optimization objective explicitly integrates attribute acquisition costs.

Definition 2 (Label-Induced Fuzzy Decision Sets)**.** *For each label $y_k \in \mathcal{Y}$, define a fuzzy decision set D_k on O via the membership function*

$$\mu_{D_k}(o_i) = \ell_i^k \quad (i = 1, \ldots, n;\ k = 1, \ldots, q). \tag{3.3}$$

The collection $\mathcal{D}_L = \{D_1, \ldots, D_q\}$ constitutes the family of label-induced fuzzy decision sets determined by L.

According to Eq. (3.1), the membership functions satisfy

$$\sum_{k=1}^{q} \mu_{D_k}(o_i) = 1 \quad \text{for all } o_i \in O. \tag{3.4}$$

Consequently, $\mathcal{D}_L$ forms a fuzzy partition-of-unity over O. Unlike traditional decision tables, $\mathcal{D}_L$ does not induce mutually disjoint crisp equivalence classes; instead, an object o_i may simultaneously hold positive membership degrees across multiple fuzzy decision sets D_k. Therefore, theoretical properties established for classical crisp partitions cannot be assumed a priori and must be formally re-established within this soft decision space.

To support the label-weighted statistics introduced in subsequent sections, the total description mass of label y_k is defined as

$$M_k = \sum_{i=1}^{n} \ell_i^k. \tag{3.5}$$

Ratios dividing by M_k are used only for labels with $M_k > 0$. A label satisfying $M_k = 0$ carries zero description mass across all sample instances and is thus naturally excluded from label-weighted computations.

Definition 3 (Test-Cost-Sensitive Label Distribution Information System)**.** *A test-cost-sensitive label distribution information system is defined as*

$$\mathcal{S}_{L,c} = (O, A, \mathcal{Y}, L, c),$$

where $c : A \to \mathbb{R}_{>0}$ assigns a positive acquisition cost to each attribute $a \in A$. Assuming additivity, the total cost of acquiring an attribute subset $P \subseteq A$ is

$$c(P) = \sum_{a \in P} c(a), \qquad \textit{with } c(\varnothing) = 0. \tag{3.6}$$

The strict positivity constraint $c(a) > 0$ guarantees strict monotonicity of acquisition costs under subset inclusion—that is, $c(P) < c(Q)$ whenever $P \subsetneq Q$—without imposing any structural monotonicity assumptions on the granular-ball dependency function itself.

Proposition 1 (Crisp Decision Representation as a Special Case)**.** *Suppose each row of L is a one-hot vector (i.e., $\ell_i^k \in \{0, 1\}$ for all i, k). Define a discrete decision mapping $d : O \to \mathcal{Y}$ such that $d(o_i) = y_k$ if and only*

if $\ell_i^k = 1$. *Then, each* D_k *collapses to the indicator function of the crisp decision class*

$$C_k = \{o_i \in O : d(o_i) = y_k\}.$$

Upon removing empty sets, $\{C_k\}$ *forms a strict crisp partition of* O.

Proof. For any $o_i \in O$, there exists a unique index k such that $\ell_i^k = 1$. Thus, o_i belongs to exactly one decision class C_k, establishing that $\bigcup_{k=1}^{q} C_k = O$ and $C_k \cap C_h = \varnothing$ for all $k \neq h$. By Eq. (3.3), the membership function reduces to

$$\mu_{D_k}(o_i) = \begin{cases} 1, & \text{if } o_i \in C_k, \\ 0, & \text{if } o_i \notin C_k, \end{cases}$$

which matches the indicator representation of the crisp set C_k. □

Proposition 1 pertains exclusively to the decision representation mapping. Whether the corresponding approximation operators and dependency metrics collapse to classical crisp rough set formulations remains a distinct theoretical question, which must be formally verified once these operators are explicitly defined. Consequently, no automatic inheritance of classical properties—such as monotonic dependency—is assumed herein.

4. Label-Distribution-Based Information Weights and Directed Dissimilarity

4.1. Local KL-Divergence Weights and Directed Dissimilarity

We use source-dependent, fixed-normalization directed aggregation with a local information statistic for numerical and categorical attributes. This statistic quantifies the divergence between an attribute-induced local label profile and the global label profile. Such a modeling choice reflects an information-theoretic design rather than an empirical claim of absolute predictive optimality.

All statistical properties are evaluated exclusively on the fixed training object set O, remaining invariant when comparing alternative attribute subsets $P, Q \subseteq A$. Validation and test labels are strictly excluded from these computations.

4.1.1. Global and Attribute-Induced Local Label Profiles

We define the global mean description degree for label y_k across training instances as

$$\bar{\ell}^k = \frac{1}{n}\sum_{i=1}^{n} \ell_i^k, \qquad \mathcal{K}_+ = \{k \in \{1,\dots,q\} : \bar{\ell}^k > 0\}. \tag{4.1}$$

The vector $\bar{\boldsymbol{\ell}} = (\bar{\ell}^1, \dots, \bar{\ell}^q)$ represents the global label distribution. By non-negativity constraint (3.1), $\bar{\ell}^k = 0$ implies $\ell_i^k = 0$ for all training instances $o_i \in O$. Such uninformative labels are naturally excluded from information ratios without altering non-zero description degrees.

For numerical attributes $a \in A_n$, training values are normalized to $[0, 1]$ via min-max scaling:

$$z_a(o_i) = \begin{cases} \dfrac{a(o_i) - a_{\min}}{a_{\max} - a_{\min}}, & \text{if } a_{\max} > a_{\min}, \\ 0, & \text{if } a_{\max} = a_{\min}, \end{cases} \tag{4.2}$$

where $a_{\min}$ and $a_{\max}$ denote the minimum and maximum values of attribute a on O. Given a fixed kernel bandwidth $\tau_a > 0$, attribute-level pairwise affinities are defined as

$$K_a(i,j) = \begin{cases} \exp\left(-\dfrac{(z_a(o_i) - z_a(o_j))^2}{2\tau_a^2}\right), & \text{if } a \in A_n, \\ 1, & \text{if } a \in A_c \text{ and } a(o_i) = a(o_j), \\ 0, & \text{if } a \in A_c \text{ and } a(o_i) \neq a(o_j). \end{cases} \tag{4.3}$$

For the experiments, let $H_a = \{|z_a(o_i) - z_a(o_j)| : i < j,\ z_a(o_i) \neq z_a(o_j)\}$. The bandwidth is $\tau_a = \text{median}(H_a)$ when H_a is nonempty, and $\tau_a = 1$ otherwise. It is fitted once on the relevant training partition; the three-object triangle-inequality example below uses its separately stated bandwidth.

For both feature modalities, the affinity matrix satisfies $0 \leq K_a(i,j) \leq 1$, $K_a(i,j) = K_a(j,i)$, and $K_a(i,i) = 1$.

We define the local support mass $s_a(i)$, the off-diagonal support mass $t_a(i)$, and the raw local label profile $\boldsymbol{u}_a(i)$ as follows:

$$s_a(i) = \sum_{j=1}^{n} K_a(i,j), \qquad t_a(i) = \sum_{j \neq i} K_a(i,j) = s_a(i) - 1, \tag{4.4}$$

$$u_a^k(i) = \frac{\sum_{j=1}^{n} K_a(i,j)\ell_j^k}{s_a(i)}. \tag{4.5}$$

Note that $s_a(i) \geq 1$ and $t_a(i) \geq 0$, ensuring $\boldsymbol{u}_a(i) = (u_a^1(i), \ldots, u_a^q(i))$ remains a valid label distribution. For categorical attributes, $\boldsymbol{u}_a(i)$ yields the conditional mean distribution for category $a(o_i)$; for numerical attributes, it computes a Gaussian kernel-weighted mean label profile around $z_a(o_i)$.

Because the raw profile includes the self-label $\boldsymbol{\ell}(o_i)$, small or sparse neighborhoods can introduce high variance. To stabilize local profile estimates, we incorporate a shrinkage hyperparameter $\beta > 0$ to construct a regularized local label profile:

$$\eta_a(i) = \frac{t_a(i)}{t_a(i) + \beta}, \tag{4.6}$$

$$\widetilde{\boldsymbol{u}}_a(i) = \eta_a(i)\boldsymbol{u}_a(i) + (1 - \eta_a(i))\bar{\boldsymbol{\ell}}. \tag{4.7}$$

The confidence factor $\eta_a(i)$ depends on off-diagonal support, although the raw profile still includes the self-label. If an object is completely isolated ($t_a(i) = 0$), $\eta_a(i) = 0$, causing the regularized profile to revert to the global profile $\bar{\boldsymbol{\ell}}$. This shrinkage mechanism balances local discrimination with sample support. Hyperparameters τ_a and β are fixed strictly during training and are not re-tuned during feature subset evaluation.

4.1.2. Normalized Local KL Divergence

When the active label set satisfies $|\mathcal{K}_+| \geq 2$, we define the dataset-level information normalization factor

$$B_L = \log_2 \left(\frac{1}{\min_{k \in \mathcal{K}_+} \bar{\ell}^k} \right) > 0. \tag{4.8}$$

The constant B_L depends solely on the fixed global training label profile and remains independent of individual attributes or selected feature subsets.

Definition 4 (Local KL-Divergence Weight)**.** *For $|\mathcal{K}_+| \geq 2$, the local KL-divergence weight of attribute a at object o_i is defined as*

$$W_a(o_i) = \frac{1}{B_L} \sum_{k \in \mathcal{K}_+} \widetilde{u}_a^k(i) \log_2 \left(\frac{\widetilde{u}_a^k(i)}{\bar{\ell}^k} \right). \tag{4.9}$$

If $|\mathcal{K}_+| = 1$, we set $W_a(o_i) = 0$ for all $a \in A$ and $o_i \in O$.

The numerator in Eq. (4.9) measures the Kullback-Leibler (KL) divergence of the regularized local profile $\widetilde{\boldsymbol{u}}_a(i)$ from the global baseline $\bar{\boldsymbol{\ell}}$ [41].

A larger weight indicates greater divergence between the local label profile induced by attribute a and the global training profile. This representation statistic does not by itself establish statistical significance or predictive discrimination.

Proposition 2 (Boundedness and Zero-Weight Characterization)**.** *For any attribute $a \in A$ and object $o_i \in O$, $0 \leq W_a(o_i) \leq 1$. Furthermore, when $|\mathcal{K}_+| \geq 2$:*

$$W_a(o_i) = 0 \quad \Longleftrightarrow \quad \widetilde{\boldsymbol{u}}_a(i) = \bar{\boldsymbol{\ell}}.$$

Specifically, $W_a(o_i) = 0$ holds if o_i lacks supporting neighbors ($t_a(i) = 0$), if a is a constant attribute, or if all training instances share identical label distributions.

Proof. Let $p_k = \bar{\ell}^k > 0$ and $r_k = \widetilde{u}_a^k(i)$ for $k \in \mathcal{K}_+$. By Gibbs' inequality (or log-sum inequality), $D_{\mathrm{KL}}(\boldsymbol{r} \parallel \boldsymbol{p}) \geq 0$, with equality holding if and only if $\boldsymbol{r} = \boldsymbol{p}$. For the upper bound,

$$D_{\mathrm{KL}}(\boldsymbol{r} \parallel \boldsymbol{p}) = \sum_{k \in \mathcal{K}_+} r_k \log_2 r_k + \sum_{k \in \mathcal{K}_+} r_k \log_2 \left(\frac{1}{p_k}\right) \leq 0 + \sum_{k \in \mathcal{K}_+} r_k B_L = B_L.$$

Dividing by B_L yields $0 \leq W_a(o_i) \leq 1$. If $t_a(i) = 0$, Eq. (4.7) gives $\widetilde{\boldsymbol{u}}_a(i) = \bar{\boldsymbol{\ell}}$, implying $W_a(o_i) = 0$. For a constant attribute, $K_a(i, j) = 1$ for all i, j, which forces $\boldsymbol{u}_a(i) = \bar{\boldsymbol{\ell}}$ and thus $W_a(o_i) = 0$. □

4.1.3. Hybrid Discrepancy and Directed Aggregation

To isolate the contribution of information weighting, we employ standard symmetric feature discrepancies. For a categorical attribute value $v \in V_a$ observed in the training sample, let

$$r_a^k(v) = \frac{\sum_{i : a(o_i) = v} \ell_i^k}{|\{i : a(o_i) = v\}|} \tag{4.10}$$

denote the average description vector for value v. The hybrid discrepancy $\Delta_a(o_i, o_j)$ is defined as

$$\Delta_a(o_i, o_j) = \begin{cases} |z_a(o_i) - z_a(o_j)|, & \text{if } a \in A_n, \\ \dfrac{1}{2} \displaystyle\sum_{k=1}^{q} |r_a^k(a(o_i)) - r_a^k(a(o_j))|, & \text{if } a \in A_c. \end{cases} \tag{4.11}$$

Both discrepancies lie in $[0,1]$, vanish on the diagonal, and satisfy symmetry and the triangle inequality. Distinct categorical values can have identical label profiles and hence zero Δ_a; this component need not separate them.

In addition to information-weighted profile discrepancy, we introduce a label-independent raw geometric discrepancy $G_a(o_i, o_j)$ to preserve topological separation for non-constant attributes that may carry zero local information weight:

$$G_a(o_i, o_j) = \begin{cases} |z_a(o_i) - z_a(o_j)|, & \text{if } a \in A_n, \\ 0, & \text{if } a \in A_c \text{ and } a(o_i) = a(o_j), \\ 1, & \text{if } a \in A_c \text{ and } a(o_i) \neq a(o_j). \end{cases} \tag{4.12}$$

Given a global mixture coefficient $\zeta \in (0,1)$, we formulate the directed dissimilarity structure as follows.

Definition 5 (Information-Weighted Directed Dissimilarity)**.** *For $m = |A| > 0$ and any subset $P \subseteq A$, the forward and backward directed dissimilarities between o_i and o_j are defined by*

$$I_P(o_i, o_j) = \frac{1}{m} \sum_{a \in P} W_a(o_i) \Delta_a(o_i, o_j), \tag{4.13}$$

$$\overrightarrow{\rho}_P(o_i, o_j) = (1 - \zeta) I_P(o_i, o_j) + \frac{\zeta}{m} \sum_{a \in P} G_a(o_i, o_j), \tag{4.14}$$

$$\overleftarrow{\rho}_P(o_i, o_j) = \overrightarrow{\rho}_P(o_j, o_i). \tag{4.15}$$

By convention, $\overrightarrow{\rho}_\varnothing(o_i, o_j) = 0$.

Proposition 3 (Properties of the Directed Dissimilarity)**.** *For any $P \subseteq Q \subseteq A$:*

1. *Boundedness: $0 \leq \overrightarrow{\rho}_P(o_i, o_j) \leq \frac{|P|}{m}$, with $\overrightarrow{\rho}_P(o_i, o_i) = 0$.*
2. *Strict Positivity: $\overrightarrow{\rho}_P(o_i, o_j) > 0$ if objects o_i, o_j differ on at least one selected numerical or categorical attribute in P.*
3. *Asymmetry: The directional asymmetry is given by*

$$\overrightarrow{\rho}_P(o_i, o_j) - \overleftarrow{\rho}_P(o_i, o_j) = \frac{1 - \zeta}{m} \sum_{a \in P} \left(W_a(o_i) - W_a(o_j) \right) \Delta_a(o_i, o_j).$$

Symmetry holds if $W_a(o_i) = W_a(o_j)$ for all $a \in P$, but equal label vectors $\boldsymbol{\ell}(o_i) = \boldsymbol{\ell}(o_j)$ are not required.

4. *Monotonicity under Inclusion:* $\overrightarrow{\rho}_P(o_i, o_j) \leq \overrightarrow{\rho}_Q(o_i, o_j)$ *for fixed statistics and dataset normalization.*

5. *Restricted Triangle Inequality: The triangle inequality holds on triples* o_i, o_j, o_k *whenever* $W_a(o_i) = W_a(o_j) = W_a(o_k)$ *for all* $a \in P$.

Proof. Boundedness follows from $W_a(o_i) \in [0,1]$ and $\Delta_a, G_a \in [0,1]$. If a selected attribute differs, its G_a term is positive; $\zeta > 0$ therefore gives strict positivity. For asymmetry, the geometric term $\frac{\zeta}{m}\sum G_a(o_i, o_j)$ is symmetric and cancels out. Monotonicity follows because appending attributes adds non-negative terms under a fixed denominator m. The restricted triangle inequality holds because both Δ_a and G_a satisfy metric triangle inequalities. □

Remark 1 (Failure of the General Triangle Inequality)**.** *The directed dissimilarity* $\overrightarrow{\rho}_P$ *does not generally satisfy the triangle inequality. Consider three objects with a single numerical attribute* $z_a(o_1) = 0, z_a(o_2) = 0.5, z_a(o_3) = 1.0$ *and label vectors* $\boldsymbol{\ell}(o_1) = (1,0), \boldsymbol{\ell}(o_2) = (0.5, 0.5), \boldsymbol{\ell}(o_3) = (0,1)$. *Let* $\tau_a = 0.5, \beta = 1$. *By symmetry,* $\widetilde{\boldsymbol{u}}_a(2) = \bar{\boldsymbol{\ell}} = (0.5, 0.5)$, *yielding* $W_a(o_2) = 0$, *whereas* $W_a(o_1) > 0$. *Evaluating forward distances gives:*

$$\overrightarrow{\rho}(o_1, o_3) = (1-\zeta)W_a(o_1) + \zeta,$$
$$\overrightarrow{\rho}(o_1, o_2) + \overrightarrow{\rho}(o_2, o_3) = \frac{1}{2}(1-\zeta)W_a(o_1) + \zeta.$$

Here, $\overrightarrow{\rho}(o_1, o_3) > \overrightarrow{\rho}(o_1, o_2) + \overrightarrow{\rho}(o_2, o_3)$ *for any* $\zeta \in (0,1)$, *showing that the directed dissimilarity* $\overrightarrow{\rho}_P$ *does not generally satisfy the triangle inequality.*

5. Distribution-Consistent Granular-Ball Representation

5.1. Label Distribution Consistency and Granular-Ball Dependency

Here, we formalize attribute-induced granular representations for LDL and derive an explained-variance dependency metric subject to support constraints. The minimum-support condition excludes singleton eligible leaves. Because eligibility and reconstruction both use training labels, it alone does not establish out-of-sample generalization.

5.1.1. Granular-Ball Representation under an Attribute Subset

Let $O = \{o_1, \ldots, o_n\}$ denote the set of training instances. For any attribute subset $P \subseteq A$, a deterministic granulation mechanism partitions O into a collection of mutually disjoint, nonempty granular balls denoted by $\mathcal{B}_P$. Because the underlying feature space is governed by the directed dissimilarity introduced in Section 4.1 rather than a standard metric, each granular ball $B \in \mathcal{B}_P$ is characterized by a triplet comprising its constituent instance subset B, a medoid representative $c_P(B)$, and a radius descriptor $r_P(B)$.

The representative $c_P(B)$ is chosen as the medoid minimizing aggregate symmetric dissimilarity:

$$c_P(B) = \operatorname*{argmin}_{o \in B} \sum_{o' \in B} d_P(o, o'), \qquad d_P(o, o') = \max\left\{ \overrightarrow{\rho}_P(o, o'), \overrightarrow{\rho}_P(o', o) \right\}, \tag{5.1}$$

where ties are deterministically resolved by the canonical index ordering of O. Although d_P does not necessarily satisfy the triangle inequality, it preserves the information-weighted directed deviations in both directions. The directional radii and the combined radius are given by

$$r_P^{\rightarrow}(B) = \max_{o \in B} \overrightarrow{\rho}_P(c_P(B), o), \qquad r_P^{\leftarrow}(B) = \max_{o \in B} \overrightarrow{\rho}_P(o, c_P(B)), \tag{5.2}$$

$$r_P(B) = \max\left\{ r_P^{\rightarrow}(B), r_P^{\leftarrow}(B) \right\}. \tag{5.3}$$

By convention, setting $P = \varnothing$ yields the trivial partition $\mathcal{B}_\varnothing = \{O\}$. Furthermore, instances that are identical across all selected attributes P are treated as an indivisible unit; label distributions alone cannot fracture attribute-indistinguishable instances.

5.1.2. Within-Ball Label Distribution Mean, Dispersion, and Consistency

Let $q = |\mathcal{Y}|$ be the number of labels, $n_B = |B|$ the ball cardinality, and $\boldsymbol{p} = \bar{\boldsymbol{\ell}} = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{\ell}_i$ the global mean label distribution across the training set. For any ball B, we define its mean label distribution $\boldsymbol{\mu}_B$ and within-ball dispersion $V(B)$ as

$$\boldsymbol{\mu}_B = \frac{1}{n_B} \sum_{o_i \in B} \boldsymbol{\ell}_i, \tag{5.4}$$

$$V(B) = \frac{1}{n_B} \sum_{o_i \in B} \|\boldsymbol{\ell}_i - \boldsymbol{\mu}_B\|_2^2, \qquad \mathrm{Cons}(B) = 1 - V(B). \tag{5.5}$$

Since each $\boldsymbol{\ell}_i \in \Delta^{q-1}$ resides on the probability simplex, the center $\boldsymbol{\mu}_B$ is also a valid probability vector. The squared Euclidean loss in label space ensures an exact ANOVA-style variance decomposition.

Proposition 4 (Properties of Within-Ball Label Distribution Dispersion)**.** *For any granular ball B, the dispersion satisfies $0 \leq V(B) \leq 1 - 1/q$, and the consistency score satisfies $1/q \leq \mathrm{Cons}(B) \leq 1$. Specifically:*

(i) $\mathrm{Cons}(B) = 1$ *(or $V(B) = 0$) if and only if all instances in B share an identical label distribution.*

(ii) If all $\boldsymbol{\ell}_i$ are degenerate one-hot distributions, $V(B)$ reduces to the standard empirical Gini impurity: $1 - \sum_{k=1}^{q} (\mu_B^k)^2$.

Proof. Expanding the squared distance around $\boldsymbol{\mu}_B$ yields

$$V(B) = \frac{1}{n_B} \sum_{i \in B} \|\boldsymbol{\ell}_i\|_2^2 - \|\boldsymbol{\mu}_B\|_2^2.$$

Because $\boldsymbol{\ell}_i \in \Delta^{q-1}$, we have $\|\boldsymbol{\ell}_i\|_2^2 \leq 1$. By the Cauchy–Schwarz inequality, $\|\boldsymbol{\mu}_B\|_2^2 \geq 1/q$. Hence, $0 \leq V(B) \leq 1 - 1/q$. The nonnegativity of squared norms implies that $V(B) = 0$ if and only if $\boldsymbol{\ell}_i = \boldsymbol{\mu}_B$ for all $o_i \in B$. When $\boldsymbol{\ell}_i$ are one-hot vectors, $\|\boldsymbol{\ell}_i\|_2^2 = 1$, which immediately yields the Gini impurity formulation. □

Proposition 4 highlights that dispersion measures distribution heterogeneity rather than informational entropy: a high-entropy but identical distribution (e.g., uniform $\boldsymbol{\ell}_i = (1/q, \ldots, 1/q)$) achieves perfect consistency ($V(B) = 0$).

To set an adaptive consistency threshold, compute the global label dispersion

$$V_0 = \frac{1}{n} \sum_{i=1}^{n} \|\boldsymbol{\ell}_i - \boldsymbol{p}\|_2^2. \tag{5.6}$$

Given an integer minimum support $s_{\min} \geq 2$, a radius bound $\alpha \in [0, 1]$, and a relative tolerance $\kappa \in [0, 1)$, we define the relative consistency threshold as $\delta = \kappa V_0$. Both V_0 and δ are determined strictly on the training partition. A ball B is termed *eligible* if it jointly satisfies support, consistency, and compactness constraints:

$$\begin{aligned} \mathrm{Eligible}_P(B) \iff n_B \geq s_{\min} \ \wedge \ V(B) \leq \delta \ \wedge \ r_P(B) \leq \alpha, \\ \mathcal{E}_P = \{B \in \mathcal{B}_P : \mathrm{Eligible}_P(B)\}. \end{aligned} \tag{5.7}$$

The predicate is defined for any nonempty candidate node, before the terminal leaf set is known. It excludes singleton eligible leaves but does not remove self-label fitting. With the experimental setting $\alpha = 1$, the radius eligibility condition is inactive because $r_P(B) \le |P|/m \le 1$. Geometry can still affect medoids, seeds and the partition.

5.1.3. Granular-Ball Dependency Based on Explained Variance

To evaluate the representative capacity of $\mathcal{B}_P$, instances belonging to eligible balls are represented by their local centers $\boldsymbol{\mu}_B$, whereas instances in ineligible balls fall back to the global prior $\boldsymbol{p}$. The resulting training reconstruction error is

$$R(P) = \frac{1}{n}\left[\sum_{B\in\mathcal{E}_P}\sum_{i\in B}\|\boldsymbol{\ell}_i - \boldsymbol{\mu}_B\|_2^2 + \sum_{B\in\mathcal{B}_P\setminus\mathcal{E}_P}\sum_{i\in B}\|\boldsymbol{\ell}_i - \boldsymbol{p}\|_2^2\right]. \tag{5.8}$$

Definition 6 (Support-Constrained Granular-Ball Dependency)**.** *The granular-ball dependency of attribute subset P is defined as the fraction of global label variance explained by the eligible balls:*

$$\Gamma(P) = \begin{cases} 1 - \dfrac{R(P)}{V_0} = \dfrac{\sum_{B\in\mathcal{E}_P} n_B\|\boldsymbol{\mu}_B - \boldsymbol{p}\|_2^2}{nV_0}, & V_0 > 0, \\ 0, & V_0 = 0. \end{cases} \tag{5.9}$$

Proposition 5 (Variance Decomposition and Boundedness)**.** *For any attribute subset $P \subseteq A$, $0 \le \Gamma(P) \le 1$ and $\Gamma(\varnothing) = 0$. For $V_0 > 0$, specifically:*

(i) $\Gamma(P) = 0$ *if and only if either* $\mathcal{E}_P = \varnothing$ *or* $\boldsymbol{\mu}_B = \boldsymbol{p}$ *for all* $B \in \mathcal{E}_P$.

(ii) $\Gamma(P) = 1$ *if and only if each eligible ball has zero dispersion ($V(B) = 0$), and every object outside the eligible balls carries exactly the global prior* $\boldsymbol{p}$.

Proof. Because $\sum_{i\in B}(\boldsymbol{\ell}_i - \boldsymbol{\mu}_B) = \mathbf{0}$, the cross-term vanishes, providing an orthogonal sum-of-squares decomposition for each ball B:

$$\sum_{i\in B}\|\boldsymbol{\ell}_i - \boldsymbol{p}\|_2^2 = n_B V(B) + n_B\|\boldsymbol{\mu}_B - \boldsymbol{p}\|_2^2. \tag{5.10}$$

Summing over all $B \in \mathcal{B}_P$ and subtracting $nR(P)$ from nV_0 cancels the intra-ball dispersion terms for eligible balls, leaving exactly the numerator in Eq. (5.9). Since $0 \le R(P) \le V_0$, the bounds follow immediately. □

5.1.4. Nonmonotonicity under Attribute-Subset Changes

A critical property of $\Gamma(P)$ is that it does not generally satisfy subset monotonicity, i.e., $P \subseteq Q \not\Longrightarrow \Gamma(P) \leq \Gamma(Q)$. Although the unconstrained between-group variation $\sum_{B \in \mathcal{B}_P} n_B \|\boldsymbol{\mu}_B - \boldsymbol{p}\|_2^2$ increases monotonically under nested partition refinements, the eligibility conditions and regenerated partitions create discrete changes:

- Regenerated partitions need not be nested; consistency and radius eligibility can change even though every accepted split respects minimum support.
- Transitioning from P to Q changes the subset-dependent dissimilarity structure, shifting representative medoids and ball boundaries nonlocally.

Consequently, attribute selection guided by $\Gamma(P)$ requires search procedures capable of handling non-monotonic objective landscapes.

5.2. Deterministic Granular-Ball Construction

To materialize the partition $\mathcal{B}_P$ deterministically, we provide a top-down, supervised binary granulation procedure.

5.2.1. Attribute-Indiscernibility Classes and Seed-Based Splitting

To preserve attribute indistinguishability, instances sharing identical feature projections on P are grouped into an *attribute-indiscernibility class* $G \subseteq B$. Each class is treated as an indivisible entity and indexed by its lowest-index instance u_G. Let $\mathcal{A}_P(B)$ denote the set of classes within B.

If $|\mathcal{A}_P(B)| \geq 2$, two distinct seed-class representatives (u, v) are identified by maximizing the mutual symmetric dissimilarity:

$$(u, v) = \underset{u_G, u_H : G \neq H}{\operatorname{argmax}} \, d_P(u_G, u_H), \quad \text{subject to } u < v, \tag{5.11}$$

with ties resolved by lexicographical index order. All classes in B are then assigned to the nearer seed:

$$B_1 = \bigcup_{G : \, d_P(u_G, u) \leq d_P(u_G, v)} G, \qquad B_2 = B \setminus B_1. \tag{5.12}$$

Tied distances are assigned to the primary seed u. Objects with identical projections on P have identical affinity rows and weights for every selected attribute, so the class representative gives the same selected-attribute dissimilarities as any member.

5.2.2. Splitting and Stopping Criteria

Given fixed nonnegative gain thresholds $\eta_V, \eta_R \geq 0$, a prospective split $(B \to B_1, B_2)$ is accepted if and only if it satisfies both conditions:

1. **Support admissibility:** $\min\{n_{B_1}, n_{B_2}\} \geq s_{\min}$.

2. **Progressive gain:** The candidate achieves a minimal reduction in either label dispersion or geometric radius:

$$g_V(B; B_1, B_2) > \eta_V \quad \text{or} \quad g_r(B; B_1, B_2) > \eta_R, \tag{5.13}$$

where the variance gain is defined as:

$$\begin{aligned} g_V(B; B_1, B_2) &= V(B) - \frac{n_{B_1} V(B_1) + n_{B_2} V(B_2)}{n_B} \\ &= \frac{n_{B_1} \|\boldsymbol{\mu}_{B_1} - \boldsymbol{\mu}_B\|_2^2 + n_{B_2} \|\boldsymbol{\mu}_{B_2} - \boldsymbol{\mu}_B\|_2^2}{n_B} \geq 0, \end{aligned} \tag{5.14}$$

and the radius contraction gain is $g_r = r_P(B) - \max\{r_P(B_1), r_P(B_2)\}$.

If a node satisfies $\text{Eligible}_P(B)$, contains fewer than $2s_{\min}$ instances, or fails both gain thresholds, it becomes a terminal leaf. The overall procedure is summarized in Algorithm 1; Figure 2 illustrates how the induced ball structure and its dependency can change non-monotonically when the attribute subset grows, which is the structural reason dependency gains must be treated as signed quantities.

Figure 2 illustrates non-monotone granulation on a synthetic eight-object construction: adding attribute b to subset $\{a\}$ reshapes the ball partition (objects re-assigned across balls, eligibility changes) and can *lower* the granular-ball dependency despite distance monotonicity. Shading distinguishes eligible from ineligible ball membership. The data are regenerated by the frozen construction with random seed 20260930 (trial 3), giving $\Gamma(\{a\}) = 0.0922402760947$ and $\Gamma(\{a, b\}) = 0$. The source project includes `figures/negative_gain_construction.csv` and `check_synthetic_example.py` for checking these values against the frozen implementation.

Proposition 6 (Termination and Determinism)**.** *Algorithm 1 terminates in at most* $\max\{0, \lfloor n/s_{\min} \rfloor - 1\}$ *splits, outputting a disjoint partition of* O *that preserves attribute-indiscernibility classes.*

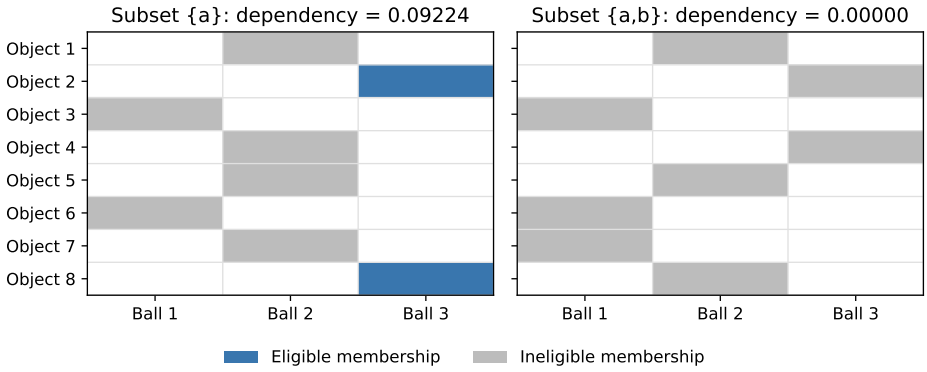


Figure 2: Nonmonotone base granular-ball dependency for eight synthetic objects. Blue denotes eligible leaves and gray denotes ineligible leaves. The two categorical attributes share the fixed denominator $m = 2$; $s_{\min} = 2$, $\kappa = 0.1$, $\alpha = 1$, $\beta = 1$, and $\zeta = 0.1$. The source project includes the generating data and a deterministic check.

Proof. If at least one split occurs, each terminal descendant has size at least $s_{\min}$, so the number of leaves L satisfies $Ls_{\min} \leq n$ and the number of splits is at most $\lfloor n/s_{\min} \rfloor - 1$. If no split occurs, the split count is zero. Determinism follows directly from the invariant tie-breaking rules on instance indices and the deterministic FIFO queue processing. □

5.2.3. Computational Complexity

Given fitted attribute relations, constructing the subset dissimilarity matrix costs $O(n^2|P|)$; projection keys are formed once and reused. A node containing b objects requires $O(b^2 + bq)$ work for the seed pair, medoids and label statistics. Summing over an unbalanced tree gives the conservative base-construction bound $O(n^3 + n^2 q)$. For balanced splits, the squared node sizes sum to $O(n^2)$ and the label-statistic work is $O(nq[1 + \log(n/s_{\min})])$, yielding $O(n^2|P| + n^2 + nq[1 + \log(n/s_{\min})])$ including geometry aggregation. These bounds assume reusable projection keys and constant-time key lookup.

Algorithm 1 Deterministic Granular-Ball Partitioning for Subset P

Require: Training labels, fixed subset geometry d_P, projection keys, and $(s_{\min}, \kappa, \alpha, \eta_V, \eta_R)$

Ensure: Granular partition $\mathcal{B}_P$, eligible ball set $\mathcal{E}_P$, dependency $\Gamma(P)$

Compute V_0 via Eq. (5.6) and set $\delta \leftarrow \kappa V_0$
Initialize empty leaf set $\mathcal{B}_P \leftarrow \emptyset$, FIFO queue $\mathcal{Q} \leftarrow [O]$
while $\mathcal{Q} \neq \emptyset$ **do**
    Dequeue current member set $B \leftarrow \mathrm{pop}(\mathcal{Q})$
    if $\mathrm{Eligible}_P(B)$ **or** $n_B < 2s_{\min}$ **then**
        $\mathcal{B}_P \leftarrow \mathcal{B}_P \cup \{B\}$; **continue**
    end if
    Form attribute-indiscernibility classes $\mathcal{A}_P(B)$
    if $|\mathcal{A}_P(B)| < 2$ **or** $\max_{G \neq H} d_P(u_G, u_H) = 0$ **then**
        $\mathcal{B}_P \leftarrow \mathcal{B}_P \cup \{B\}$; **continue**
    end if
    Find seed pair (u, v) via Eq. (5.11) and split into (B_1, B_2) via Eq. (5.12)
    if $\min\{n_{B_1}, n_{B_2}\} \geq s_{\min}$ **and** $(g_V > \eta_V$ **or** $g_r > \eta_R)$ **then**
        Enqueue $\mathcal{Q} \leftarrow \mathrm{push}(\mathcal{Q}, B_1, B_2)$
    else
        $\mathcal{B}_P \leftarrow \mathcal{B}_P \cup \{B\}$
    end if
end while
Extract $\mathcal{E}_P \leftarrow \{B \in \mathcal{B}_P : n_B \geq s_{\min} \wedge V(B) \leq \delta \wedge r_P(B) \leq \alpha\}$
Compute $\Gamma(P)$ via Eq. (5.9)
return $\mathcal{B}_P$, $\mathcal{E}_P$, $\Gamma(P)$

6. Objectwise-Preserving Cost-Sensitive Reduction

Figure 3 summarizes the fixed-reference search procedure of Algorithm 3. Throughout the search, the training-fitted profiles, weights, and original denominator m remain invariant, whereas each candidate subset regenerates its granular balls and anchored reconstruction. The acceptance test is consistently evaluated against the initial full-attribute error vector. A successful attribute deletion does not trigger a restart of the current sweep; instead, previously rejected attributes are reconsidered in a subsequent complete sweep. Note that the flowchart illustrates the nondegenerate-reference regime. As

detailed below, the boundary case $V_0 = 0$ terminates with an `Empty` output, whereas $V_0 > 0$ with zero anchored full-set dependency triggers the predeclared exclusion described below. The final nonempty subset is guaranteed to be single-deletion irreducible, rather than representing the outcome of a global minimum-cost search.

In the flowchart, "fit and fix geometry" retains the training-fitted profiles, weights and denominator m. Each candidate reconstruction regenerates the granular balls and anchored reconstruction before evaluating its errors. The sweep order is descending cost with index-based tie breaking, and its flag records whether any deletion was accepted. An unsuccessful candidate leaves P unchanged. A complete sweep with no accepted deletion yields the local stopping certificate; an empty subset terminates directly.

6.1. Anchored Extension of the Training Representation

Let $\mathcal{E}_P^0$ be the eligible leaves from Algorithm 1. Their base reconstruction is $\boldsymbol{\mu}_B$ on those leaves and $\boldsymbol{p}$ elsewhere. The anchored oracle first computes this base construction, then runs a bounded alternative-seed construction from the root O with the same eligibility, assignment, support and gain rules. For each nonterminal node, all positive-distance pairs of distinct class representatives are ordered by decreasing $d_P(u, v)$, then increasing (u, v) to break exact ties. The budget K includes the first, farthest pair. The first pair satisfying support and progressive gain is used; its children need not yet be eligible. If no pair among the first K passes, the node is terminal. The queue is FIFO, and child members are kept in original-index order.

This procedure reproduces every successful base split before considering alternatives at a node where the base procedure stopped. Induction down the base tree therefore preserves every base eligible leaf. Alternative splitting occurs only within base-ineligible regions; it never merges leaves or splits an eligible anchor. A newly eligible leaf is accepted as a whole only if

$$\|\boldsymbol{\ell}_i - \boldsymbol{\mu}_B\|_2^2 \le \|\boldsymbol{\ell}_i - \boldsymbol{p}\|_2^2 + \epsilon_{\text{safe}} \quad \text{for every } i \in B, \qquad \epsilon_{\text{safe}} \ge 0. \tag{6.1}$$

An unsafe new leaf is not split again after this final safety test: all its members retain the global fallback. Original eligible leaves are retained without imposing a new safety test. With $\mathcal{L}_P^{(K)}$ denoting the alternative terminal partition, the disjoint accepted leaves are

$$\widetilde{\mathcal{E}}_P = \mathcal{E}_P^0 \ \cup \ \{B \in \mathcal{L}_P^{(K)} : \text{Eligible}_P(B),\ B \notin \mathcal{E}_P^0,\ B \text{ satisfies Eq. (6.1)}\}.$$

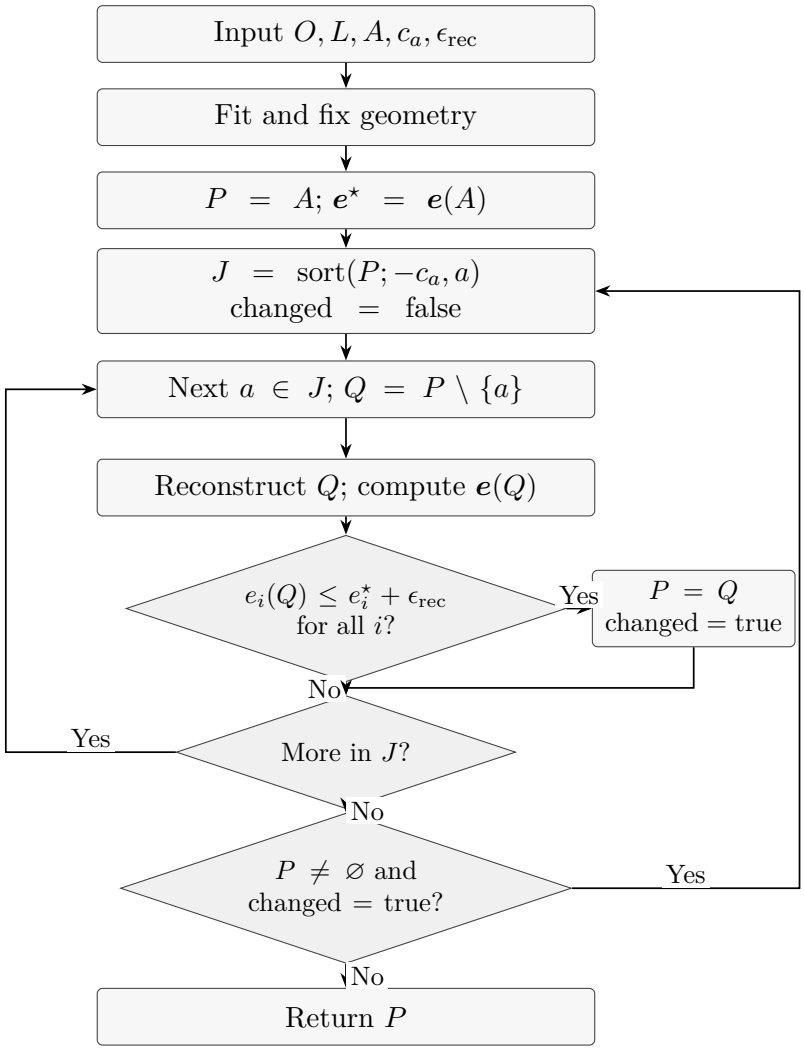


Figure 3: GBIP-O1 deletion workflow for $V_0 > 0$ and $\widetilde{\Gamma}(A) > 0$. Every candidate is compared with the same full-set error vector. Successful deletions continue the current sweep; zero-reference status handling is specified in the text.

The anchored reconstruction and its error are

$$\widetilde{\boldsymbol{\ell}}_i(P) = \begin{cases} \boldsymbol{\mu}_B, & i \in B \in \widetilde{\mathcal{E}}_P, \\ \boldsymbol{p}, & \text{otherwise}, \end{cases} \qquad e_i(P) = \|\boldsymbol{\ell}_i - \widetilde{\boldsymbol{\ell}}_i(P)\|_2^2. \tag{6.2}$$

Define the anchored dependency explicitly by

$$\widetilde{\Gamma}(P) = \begin{cases} 1 - \dfrac{n^{-1}\sum_i e_i(P)}{V_0}, & V_0 > 0, \\ 0, & V_0 = 0. \end{cases} \tag{6.3}$$

The same per-leaf variance decomposition as for Γ gives $0 \le \widetilde{\Gamma}(P) \le 1$. Algorithm 2 specifies the oracle; no test labels enter it.

Proposition 7 (Within-subset safety)**.** *For a fixed subset P, the anchored extension satisfies $e_i(P) \le e_i^0(P) + \epsilon_{\text{safe}}$ for every object, where $e_i^0(P)$ is its base reconstruction error. For $V_0 > 0$, $\widetilde{\Gamma}(P) \ge \Gamma(P) - \epsilon_{\text{safe}}/V_0$; in exact arithmetic with $\epsilon_{\text{safe}} = 0$, $\widetilde{\Gamma}(P) \ge \Gamma(P)$.*

Proof. Retained base eligible leaves keep their reconstructions. A newly accepted leaf lies in a region that previously used $\boldsymbol{p}$, and satisfies Eq. (6.1) for every member. All other objects retain the fallback. These cases cover O. Averaging the inequalities and applying Eq. (6.3) gives the dependency bound. □

This property compares two constructions of the same subset. Preservation across different subsets is imposed separately below.

6.2. Fixed-Reference Preservation Objective

Let $c_a = c(a) > 0$ and $C(P) = c(P) = \sum_{a \in P} c_a$. Fit all geometry statistics on the training sample, retain the original full attribute denominator m, and fix $e_i^\star = e_i(A)$. The admissible family is

$$\mathcal{F}_{\text{O1}} = \{P \subseteq A : e_i(P) \le e_i^\star + \epsilon_{\text{rec}} \text{ for all } i\}, \qquad P^\star \in \operatorname*{argmin}_{P \in \mathcal{F}_{\text{O1}}} C(P). \tag{6.4}$$

The error defined in Eq. (6.2) represents the unnormalized sum of squared label-component errors. The benchmark vector $\boldsymbol{e}^\star$ is computed once prior to the search procedure and remains fixed, rather than being updated by the errors of subsequently evaluated subsets. While attribute-local profiles,

Algorithm 2 Anchored training reconstruction for subset P

Require: Fixed d_P, labels L, projection keys, representation parameters, $K \in \mathbb{N}_{>0}$, $\epsilon_{\text{safe}} \geq 0$

Run Algorithm 1; retain its eligible anchors $\mathcal{E}_P^0$
$\mathcal{L} \leftarrow \varnothing$; initialize FIFO queue $\mathcal{Q} \leftarrow [O]$
while $\mathcal{Q} \neq \varnothing$ **do**
    Pop B; compute its medoid, radius and dispersion
    if Eligible$_P(B)$ or $|B| < 2s_{\min}$ **then**
        Add B to $\mathcal{L}$; **continue**
    end if
    Form indivisible projection classes and their lowest-index representatives
    Order positive-distance representative pairs by $(-d_P(u, v), u, v)$
    split $\leftarrow$ false
    for each of the first $\min(K, \text{number of pairs})$ pairs **do**
        Assign whole classes by Eq. (5.12); sort child members by index
        if both children have size $\geq s_{\min}$ and Eq. (5.13) holds **then**
            Append both children to $\mathcal{Q}$; split $\leftarrow$ true; **break**
        end if
    end for
    if split = false **then**
        Add B to $\mathcal{L}$
    end if
end while
Verify that every anchor in $\mathcal{E}_P^0$ is a member of $\mathcal{L}$
$\widetilde{\mathcal{E}}_P \leftarrow \mathcal{E}_P^0$
for each $B \in \mathcal{L} \setminus \mathcal{E}_P^0$ **do**
    if Eligible$_P(B)$ and Eq. (6.1) holds for every member **then**
        Add the whole leaf B to $\widetilde{\mathcal{E}}_P$
    end if
end for
Compute reconstructions, $\boldsymbol{e}(P)$ and $\widetilde{\Gamma}(P)$ by Eqs. (6.2)–(6.3)
return $\boldsymbol{e}(P)$, $\widetilde{\mathcal{E}}_P$, $\widetilde{\Gamma}(P)$

information weights, and the original denominator are held constant, each candidate subset determines its specific dissimilarities and regenerates the corresponding granular balls and accepted leaves. Furthermore, the established bound controls the allowable error increase relative to the observed label distributions, rather than measuring the distance between two reconstructed distributions. For a user-defined scientific tolerance $\tau \geq 0$, the practical implementation adopts $\epsilon_{\mathrm{rec}} = \tau + \epsilon_{\mathrm{num}}$, where $\epsilon_{\mathrm{num}} = 10^{-12}$ accounts for numerical slack in our experiments. Consequently, strict preservation corresponds to $\tau = 0$ rather than implying a strict zero reconstruction error. Finally, kernel bandwidths τ_a and split-gain thresholds η_V, η_R serve as distinct representation parameters, whereas ϵ_{rec} specifically denotes the reconstruction allowance.

The full attribute set is guaranteed to be feasible. Under exact arithmetic with $\epsilon_{\mathrm{rec}} = 0$, any admissible subset maintains a mean reconstruction error not exceeding that of the reference baseline; however, the converse does not hold since averaging permits error compensation across instances. For $V_0 > 0$, the tolerance-aware formulation additionally satisfies

$$\widetilde{\Gamma}(P) \geq \widetilde{\Gamma}(A) - \frac{\epsilon_{\mathrm{rec}}}{V_0}. \tag{6.5}$$

When $V_0 = 0$, all target distributions equal $\boldsymbol{p}$ and Empty has zero error and cost. When $V_0 > 0$, the frozen implementation computes $\widetilde{\Gamma}(A) = 1 - \mathrm{mean}(\boldsymbol{e}^\star)/V_0$ and, if this value is exactly zero, records `degenerate_reference` and returns no selected subset. This is a predeclared inclusion rule for the positive-reference experiments, not a consequence of infeasibility of Eq. (6.4); A remains feasible. For example, two objects with identical attributes and labels $(0.9, 0.1)$ and $(0.1, 0.9)$ both use the global fallback, so Full and Empty have the same errors although the implementation excludes the zero-reference case. The two excluded benchmark splits retain their recorded status rather than being retrospectively replaced by Empty. No near-zero cutoff is substituted for the exact-zero implementation test.

Retaining the full feature set may stem from two distinct phenomena: either all proper subsets are infeasible, or the greedy deletion search fails to navigate toward a feasible proper subset. Consequently, rejecting all single-attribute deletions establishes solely single-deletion irreducibility; establishing the global infeasibility of compression for a given instance requires either an exhaustive search or a dedicated structural proof.

6.3. A Sufficient Condition for the Dominant Label

Proposition 8 (Target-margin certificate)**.** *Let $q \geq 2$ and suppose the target vector $\boldsymbol{\ell}_i$ has a unique maximum at label index $k_i^\star$. Define $\gamma_i = \ell_i^{k_i^\star} - \max_{b \neq k_i^\star} \ell_i^b > 0$. If*

$$e_i^\star + \epsilon_{\text{rec}} < \gamma_i^2/2, \tag{6.6}$$

then every O1-feasible reconstruction of object i, including its full-set reference, has the same unique maximum $k_i^\star$.

Proof. Write $\boldsymbol{q}$ for a reconstruction. If $q_b \geq q_{k_i^\star}$ for some $b \neq k_i^\star$, then $(\ell_i^{k_i^\star} - q_{k_i^\star}) - (\ell_i^b - q_b) \geq \gamma_i$. Cauchy–Schwarz implies $\|\boldsymbol{\ell}_i - \boldsymbol{q}\|_2^2 \geq \gamma_i^2/2$, contradicting Eq. (6.6) and the O1 bound. The full-set reference is itself O1-feasible, so the conclusion also applies to it. □

The strict inequality is necessary for this sufficient certificate: equality can permit a predicted tie. The result neither certifies all label rankings nor extends to unlabeled test objects without further assumptions.

6.4. Complete-Sweep Deletion

GBIP-O1 initializes the search from the full attribute set $P = A$. Within each iteration pass, attributes present at the beginning of the pass are evaluated in descending order of their acquisition costs, with original index positions serving as a tie-breaker. For each candidate attribute a, the reduced subset $P \setminus \{a\}$ is tested against the fixed reference error vector $\boldsymbol{e}^\star$, accepting the deletion if and only if the feasibility condition in Eq. (6.4) is satisfied. Each pass runs to completion before restarting, and the search terminates when a full pass yields no accepted deletions. The complete procedure is formalized in Algorithm 3, where previously evaluated subsets can be efficiently cached via a deterministic oracle.

Proposition 9 (Feasibility, finite termination and stopping certificate)**.** *For a nondegenerate reference, GBIP-O1 maintains feasibility and makes at most $m(m+1)/2$ deletion-candidate requests. It returns either Empty or a feasible set for which every retained-attribute deletion is infeasible.*

Proof. The full attribute set is guaranteed to be feasible, and the acceptance criterion preserves this invariant. Each accepted deletion strictly reduces both subset cardinality and total cost. A successful pass starting with s attributes evaluates at most s candidates, resulting in a subset of size at most

Algorithm 3 GBIP-O1: Complete-Sweep Deletion from the Full Attribute Set

Require: Nondegenerate full reference, positive costs c_a, deterministic training error oracle $\boldsymbol{e}(P)$, tolerance ϵ_{rec}

$P \leftarrow A$; $\boldsymbol{e}^{\star} \leftarrow \boldsymbol{e}(A)$
repeat
  changed $\leftarrow$ false; $J \leftarrow \mathrm{sort}(P; -c_a, a)$
  for each $a \in J$ **do**
    $Q \leftarrow P \setminus \{a\}$
    if $e_i(Q) \leq e_i^{\star} + \epsilon_{\mathrm{rec}}$ for every i **then**
      $P \leftarrow Q$; changed $\leftarrow$ true
    end if
  end for
until $P = \varnothing$ or changed = false
return P

$s - 1$. Consequently, the starting cardinalities across successive passes are bounded by $m, m-1, \ldots, 1$, including the final unsuccessful verification pass if present, yielding a cumulative upper bound of $m(m+1)/2$. In a terminating (unchanged) pass, the evaluated subset remains invariant throughout, and every deletion of a retained attribute is tested against this identical base; consequently, all such candidate deletions are proven infeasible upon termination. □

Single-deletion irreducibility constitutes a local certificate: under nonmonotone granulation, multiple simultaneous attribute deletions may well yield another valid feasible subset. In contrast, exhaustive enumeration over all 2^m subsets followed by a comparison of feasible costs constitutes a distinct global-optimality procedure, as no monotonicity-based subtree pruning is presupposed or relied upon herein.

6.5. Computational Requirements

For the original eager relation-cache implementation, fitting costs $\mathcal{O}(mn^2q)$ arithmetic and $\mathcal{O}(mn^2+mnq)$ numerical storage. Subset geometry aggregation costs $\mathcal{O}(n^2|P|)$. The alternative construction sorts up to $O(b^2)$ seed pairs at a node of size b, at cost $O(b^2 \log b)$, and tests up to K pairs with $O(b^2 + bq)$ work each. Including the base construction, an unbalanced-tree oracle therefore has the conservative bound $\mathcal{O}(n^2|P|+n^3 \log n+Kn^3+Kn^2q)$.

The deletion-request bound gives $\mathcal{O}(m^2[mn^2 + n^3 \log n + Kn^3 + Kn^2q])$ after fitting, plus at most m independently recomputed terminal deletion checks. The equivalent numerical streaming implementation in Section 7.7.2 instead retains normalized attributes and weights and recomputes subset relations, using $\mathcal{O}(mn + n^2 + nq)$ numerical geometry workspace. Exact top-budget pair selection changes the sorting work on the evaluated data, without removing the conservative tie-heavy worst-case bound above. Cached error vectors require additional storage depending on the queried subsets; geometry workspace alone is not a process peak-memory measurement. Acquisition cost $C(P)$ is the specified attribute budget and is distinct from this selection workload and from prediction latency.

7. Experimental Evaluation

We evaluate aggregate quality and objectwise preservation, search and geometry effects, feasibility and parameter sensitivity, downstream decisions, and measured image-pipeline resources.

7.1. Experimental Setup

We evaluate attribute reduction across 25 native label distribution learning (LDL) benchmark tasks originating from 16 distinct feature families, spanning scene annotation, facial expression analysis, image aesthetics, emotion recognition, gene expression, and biological aging. Table 1 summarizes the benchmark tasks and their empirical coverage. Notably, the ten Yeast labeling conditions share a common feature matrix and thus contribute a single unit of weight to family-level summaries; incomplete family–method combinations are designated as missing. Furthermore, M2B is reported separately as a predictor-suitability boundary case.

Specifically, Table 1 lists the benchmark tasks, source-array dimensions, and sample sizes per frozen split, where N, m, and q denote the numbers of source objects, attributes, and label dimensions, respectively. The training and test sizes pertain to the primary benchmark established across three frozen random splits per task. As noted, while the 25 main tasks draw from 16 feature families (with Yeast tasks sharing a feature space), M2B is treated independently as a dedicated boundary case.

Each dataset is partitioned into three predetermined training/test splits shared uniformly across all competing methods. Specifically, the original splits for Natural Scene, SBU-3DFE, and Yeast-cold comprise 192 training

Table 1: Benchmark Datasets and Split Sizes

Task	N	m	q	Training	Test
Natural Scene	2000	294	9	192	256
SBU-3DFE	2500	243	6	192	256
Yeast-cold	2465	24	4	192	256
Yeast-spo	2465	24	6	192	256
Yeast-alpha	2465	24	18	192	256
Yeast-cdc	2465	24	15	192	256
Yeast-diau	2465	24	7	192	256
Yeast-dtt	2465	24	4	192	256
Yeast-elu	2465	24	14	192	256
Yeast-heat	2465	24	6	192	256
Yeast-spo5	2465	24	3	192	256
Yeast-spoem	2465	24	2	192	256
FG-Net	1002	251	70	192	256
SJAFFE	213	243	6	159	54
Music	360	128	9	192	168
Painting	280	142	8	192	88
SCUT-FBP	1500	300	5	192	256
fbp5500	5500	512	5	192	256
emotion6	1980	168	7	192	256
RAF-ML	4908	200	6	192	256
Movie	7755	1869	5	192	256
Flickr	11150	168	8	192	256
Twitter	10045	168	8	192	256
MOR-PH	55132	200	62	192	256
Human-Gene	17892	36	68	192	256
M2B	1240	250	5	192	256

and 256 test instances. For the newly incorporated datasets, the training and test sizes are determined by $n_{\text{train}} = \min(192, \lfloor 0.75N \rfloor)$ and $n_{\text{test}} = \min(256, N - n_{\text{train}})$, respectively. Accordingly, SJAFFE utilizes 159/54 instances, Music 192/168, and Painting 192/88, whereas all remaining added datasets adopt the standard 192/256 split. Each split is disjoint internally, but different splits may reuse objects. The first three datasets were used in development; the added datasets broaden coverage but had also been examined under an earlier algorithm. These are not previously unseen confirmatory test collections. Most source objects outside the stated subsamples are unused in the primary benchmark; larger-training-sample diagnostics are reported separately.

For the primary benchmark, attribute acquisition costs are simulated as positive integers drawn from $\{1, \ldots, 10\}$ and held invariant across repeated runs of each dataset (with cost sensitivity examined later in Section 7.4.3). The original cost vectors are preserved for the three initial datasets, while added-task prices use NumPy's `default_rng(20260931)` and are fixed across repeats. For a new task and repeat r, the split uses seed $2026102900 + r$; test objects are the first n_{test} entries of the permutation and training objects the next n_{train}. Three inner folds partition a training-index permutation using seed $2026102900 + r + 7000$. The original nine splits are reused from their frozen index files. Labels must be finite and nonnegative with positive row sums before normalization; zero-sum rows are rejected. All finite numerical columns, including training-constant columns, are retained. Predictive preprocessing is fitted only on its fitting partition.

The cost tables report simulated budgets allocated for acquiring the selected subset of attributes. For instance, the SCUT-FBP LDL study uses 300 PCA coordinates extracted from image descriptors [4], and our simulation prices each coordinate individually. Such coordinates share a feature-extraction pipeline; dropping them does not establish an equivalent reduction in physical acquisition or front-end extraction cost.

Furthermore, dataset names may encompass varying released versions. Specifically, the initial SCUT-FBP description reports 500 subjects [42], whereas the subsequent LDL adaptation comprises 1,500 instances [4], which precisely aligns with our array row count. Similarly, while the original LDL repository documents 30,542 human-gene objects [1], our frozen experimental source contains 17,892 instances. Because row-level transformations and identity mappings across these evolutionary versions remain unverified, all reported results pertain to the recorded source arrays and frozen splits. The

light results archive provides source URLs, checksums and dimensions in `dataset_manifest.json`; it does not resolve unverified upstream row filtering. The splits are object-level, with no verified subject-disjoint protocol for datasets containing multiple observations of a subject. Generalization to new subjects is therefore not established.

GBIP-O1 fits its information geometry once per training split and initializes the reduction process from the full attribute set. All deletion evaluations uniformly adopt the invariant full-set reference error vector, setting the scientific tolerance to $\tau = 0$ with a numerical slack of $\epsilon_{\text{num}} = 10^{-12}$. The representation settings are $\beta = 1$, $\zeta = 0.1$, $s_{\min} = 2$, $\kappa = 0.1$, $\alpha = 1$, $K = 256$, $\eta_V = \eta_R = 0$ and $\epsilon_{\text{safe}} = 10^{-12}$. Numerical bandwidths follow the positive-pairwise-difference median rule in Section 4.1. The pair budget includes the farthest pair, and the new-leaf safety slack is distinct from the across-subset tolerance. At $\alpha = 1$, compactness eligibility is automatically satisfied; the distance still affects the partition. Complete cost-ordered deletion passes execute until an unchanged pass triggers termination, with each retained-attribute deletion verified independently. Predictive loss is not used in the GBIP-O1 deletion-acceptance rule; comparator parameters are selected by the training-internal procedure described below.

Comparators and Outcome Measures. The baseline comparison encompasses GBFRS, IWDLFS, and IWCLFS [9, 10]. Specifically, GBFRS-EQ denotes an equation-based reconstruction variant utilizing the disclosed neighborhood interpretation, optimized over a 200-combination grid spanning $\alpha \in \{0.05, 0.15, \ldots, 0.95\}$ and $\beta \in \{7, 9, \ldots, 45\}$. Its selected feature prefix retains the floor of 40%, 30%, or 20% of the total attributes for feature dimensions up to 100, 500, or 1,000, respectively. For the high-dimensional Movie dataset comprising 1,869 attributes, the 373-attribute output is determined via an explicitly extended 20% rule. Furthermore, the primary IW implementations restore the label-distance normalization factor prescribed in the original literature while preserving the remaining released selection logic. Each IW method selects its parameter λ from $\{0.1, 0.2, \ldots, 0.9\}$. The frozen runner records the reconstruction conventions and source hashes. GBFRS-EQ is an equation-based implementation with disclosed boundary choices, not an author-confirmed reproduction of the original complete pipeline. The IW label-distance correction and released-distance sensitivity are distinct configurations; neither is silently substituted for a failed output.

Comparator hyperparameters are selected to minimize the object-

weighted Brier error across three training-internal cross-validation folds. Both feature scaling parameters and selector statistics are refitted independently within each respective fitting fold. Exact ties favor the smaller α and then the smaller β for GBFRS-EQ, and the smaller λ for each IW method. Final hyperparameter selections are fixed prior to evaluation on a separate held-out test set. This shared downstream predictor controls the prediction stage when comparing distinct attribute subsets, although the shared protocol does not reproduce each original paper's complete evaluation pipeline. The common predictor uses training min–max scaling, with zero spans replaced by one and test coordinates left unclipped. It orders training objects by squared Euclidean distance on the selected coordinates, uses a stable index-preserving sort for ties, and averages the first five training label vectors, renormalizing to unit sum. Empty predicts the global training label mean. For the numerical-array benchmark this defines the complete prediction rule; heterogeneous-feature theory is not a claim that a categorical prediction pipeline was benchmarked. The primary Brier error is the mean over test objects of $\sum_{k=1}^{q}(\ell_i^k - \widehat{\ell}_i^k)^2$, without division by q. Supplementary TV is the mean half-L_1 distance. KL is $D_{\mathrm{KL}}(\boldsymbol{\ell}_i \| \widehat{\boldsymbol{\ell}}_i)$ with natural logarithms after clipping predicted components below 10^{-12} and renormalizing. JS averages the two natural-log KL terms to the midpoint distribution; zero target terms contribute zero and denominators in the implementation are protected at 10^{-300}. These conventions are fixed across methods.

GBFRS-EQ selects features without imposing the fixed objectwise constraint. In contrast, the common-preservation comparison explicitly enforces this constraint upon its feature ranking. For a matched-constraint comparison, we separately evaluate the least-cost prefix of the full GBFRS-EQ ranking that satisfies the identical objectwise reference condition as GBIP-O1, denoted as GBFRS+O1, with all $m + 1$ prefixes, including Empty and Full, evaluated without monotonicity pruning or additional deletion passes. Furthermore, the purity-shortcut interpretation evaluated at primary GBFRS parameters, the fixed IW outputs ($\lambda = 0.8$), and the released-distance IW variants are retained as supplementary sensitivity configurations, each reported independently.

Unavailable Outputs. The IW implementations generate nonfinite released information weights on the Painting and Movie datasets. Consequently, both methods fail to yield a valid cross-validation-selected output on these benchmarks, and their fixed-parameter configurations exhibit the same undefined

status. Coverage records distinguish `nonfinite_released_weights` or no valid CV candidate from GBIP-O1's predeclared `degenerate_reference` exclusion. Both suppress complete three-repeat means, but they have different causes. Furthermore, calculating the three-repeat means requires three valid outputs. While comprehensive supplementary tables provide every configuration, standard deviation, and operational status, M2B remains preserved as a separately identified boundary case.

Reproducibility. The public repository indexes the current manuscript, the legacy results and reviewer archives, and the additional actual-data supplements. The light 25-task results archive at `review_audit/complete25/LDL_25_results_light.zip` contains `all_results.csv`, `selected_subsets.json`, all 17,004 parameter-CV records, `protocol.json`, `dataset_manifest.json` and freeze hashes. Frozen experiment modules and original split files are embedded in `local_runner/BOOTSTRAP_25.py`; its verified payload includes `granulation_reference.py`, `information_reduction/safe_extension.py`, `information_reduction/alternative_seeds.py` and `information_reduction/formal_method.py`. The retained runner amendments document the GBFRS numerical-radius repair and exact-score optimization used in later cases.

The separate reviewer archive, linked from the repository's submission index, supports `python run_checks.py -figures`. Its six checks recompute candidate-summary audits, recorded timing and storage arithmetic, object-decision diagnostics, feature-family statistics and figures from frozen inputs. They do not rerun original-data selection or prediction, ResNet inference, new hardware timing, all error-vector hashes, or original-query-pool disjointness. The full original return archive and image-pipeline SQLite cache are not included. Passing these checks therefore supports the specified record-level computations, not complete end-to-end reproduction of every table. The newer actual-data runs are separate: `experiments/additional_20261113/` covers the 27 geometry runs and image-prediction replay, and `experiments/official_image_selection_20261113/` covers the complete 3,300-object strict-O1 selection. Their recorded protocols, code and aggregate results are public; original inputs and individual-error caches are retained privately. These runs do not establish subject-disjoint evaluation or repeat every benchmark comparator.

7.2. Aggregate Quality and Objectwise Preservation

7.2.1. Acquisition Cost and Constraint Compatibility

Table 2 summarizes the mean retained attribute cost across three frozen benchmark splits, expressed as a percentage of the full-attribute cost: $100C(P)/C(A)$, where lower values indicate lower-cost attribute subsets. Here, GBFRS+O1 denotes the least-cost feasible prefix of the GBFRS-EQ ranking evaluated under the shared evaluator. Dashes indicate unavailable three-split means, with specific reasons detailed in the protocol. Furthermore, full attribute retention is established from exact subset records rather than rounded percentage figures.

Table 2 details the resulting subset acquisition costs. Specifically, GBIP-O1 yields a valid nondegenerate-reference output in 73 out of the 75 main dataset–repeat instances, including full-retention outputs, with all 73 satisfying the specified objectwise condition. Two isolated cases—comprising one repeat each of the Yeast-alpha and Painting datasets—exhibit a degenerate full-set reference and are thus reported without a reduction output; consequently, complete three-repeat means are available for 23 main datasets.

Among these 23 evaluated datasets, 8 permit lower mean acquisition costs, whereas 15 retain the full-set cost entirely. The most substantial cost reduction is achieved on Movie (90.12%), followed by SCUT-FBP (44.91%) and SBU-3DFE (20.83%), with the remaining reductions ranging from 0.34% to 12.97%. These empirical findings reveal a strong dependence on the fitted information representation: redundant attribute acquisition is successfully removable on certain datasets, whereas the specified single-deletion search retains every attribute on others.

Although the baseline implementations evaluated here typically acquire fewer attributes, their optimization objectives differ from the specified objectwise requirement. Specifically, GBFRS-EQ satisfies this requirement in only 4 out of 75 main cases, IWDLFS in 8 out of 69 valid cases, and IWCLFS in 20 out of 69 valid cases. In contrast, the separately wrapped GBFRS+O1 output achieves 100% compliance, satisfying the condition across all 75 evaluated cases. Figure 4 summarizes these compliance counts at the split level rather than quantifying the number or severity of individual object-level violations, with GBFRS+O1 similarly enforcing the requirement (though its search costs are analyzed separately).

As illustrated in Figure 4, the compatibility of selected subsets with the fixed-reference objectwise bound is evaluated across 75 task–split instances.

Table 2: Retained Attribute Cost

	O1	Without O1			O1
Task	GBIP-O1	GBFRS-EQ	IWDLFS	IWCLFS	GBFRS+O1
Natural Scene	95.00	28.44	71.54	94.30	100.00
SBU-3DFE	79.17	29.16	87.21	98.45	99.93
Yeast-cold	100.00	36.22	3.15	35.43	100.00
Yeast-spo	100.00	35.17	64.57	52.49	100.00
Yeast-alpha	–	39.37	3.15	100.00	44.62
Yeast-cdc	100.00	38.06	3.15	35.43	100.00
Yeast-diau	100.00	37.53	97.90	52.49	81.10
Yeast-dtt	100.00	42.78	93.96	100.00	100.00
Yeast-elu	98.95	41.21	3.15	55.64	100.00
Yeast-heat	100.00	39.63	65.09	3.15	100.00
Yeast-spo5	100.00	37.01	32.02	35.43	100.00
Yeast-spoem	100.00	39.63	65.35	67.72	100.00
FG-Net	100.00	29.86	30.69	72.26	100.00
SJAFFE	100.00	29.81	82.33	95.36	100.00
Music	100.00	31.78	69.77	52.66	100.00
Painting	–	27.60	–	–	66.67
SCUT-FBP	55.09	29.86	0.24	99.84	100.00
fbp5500	93.99	19.35	89.08	90.98	100.00
emotion6	87.03	30.54	64.42	63.91	99.76
RAF-ML	100.00	31.12	0.34	0.34	100.00
Movie	9.88	20.34	–	–	99.67
Flickr	100.00	30.17	43.48	91.56	100.00
Twitter	99.66	27.24	46.41	63.53	100.00
MOR-PH	100.00	31.96	0.34	0.34	100.00
Human-Gene	100.00	36.83	34.50	90.68	100.00

A subset is deemed *feasible* if every training object successfully passes the constraint, whereas *violating* indicates that at least one instance fails. Unavailable outputs are treated independently, comprising two degenerate GBIP-O1 references and six missing outputs per IW method due to nonfinite weights. Consequently, the feasible-to-valid counts for the plotted methods in standard order are 73/73 (GBIP-O1), 4/75 (GBFRS-EQ), 8/69 (IWDLFS), 20/69 (IWCLFS), and 75/75 (GBFRS+O1).

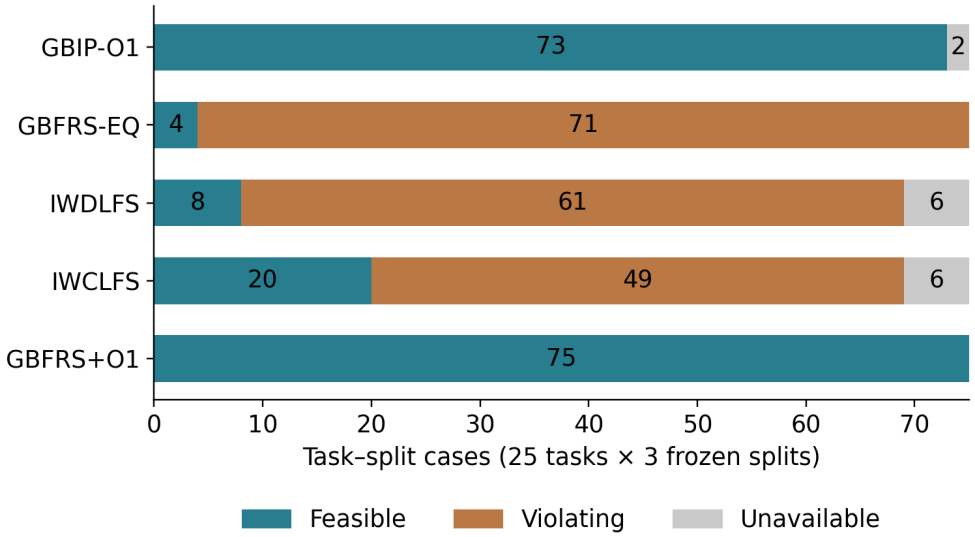


Figure 4: Split-level compatibility with the O1 training-error bound on 25 tasks and three splits each. Feasible means all training objects pass; unavailable includes separately reported zero-reference exclusions and invalid IW outputs.

7.2.2. Downstream Prediction

Table 3 reports mean test Brier error across three frozen splits under the common five-neighbor predictor; lower values indicate smaller prediction error. Full uses all attributes; Empty predicts the training label mean. Dashes denote unavailable three-split means. Comparisons use unrounded values; split results and standard deviations are supplied in the supplement.

In Table 3, across the 23 datasets with complete GBIP-O1 means, its mean Brier error is lower than Full on three datasets, higher on five, and equal on fifteen. Movie combines the largest cost saving with a decrease from 0.04862950 to 0.04386832. The remaining reduced subsets show both predictive improvements and deterioration. Figure 5 shows the cost savings

Table 3: Test Brier Error

Task	Full	Empty	GBIP-O1	GBFRS-EQ	IWDLFS	IWCLFS
Natural Scene	0.24528	0.30422	0.24675	0.23767	0.25345	0.25132
SBU-3DFE	0.03115	0.03227	0.03148	0.03068	0.03137	0.03098
Yeast-cold	0.00771	0.00655	0.00771	0.00766	0.00773	0.00767
Yeast-spo	0.00956	0.00792	0.00956	0.00947	0.00950	0.00945
Yeast-alpha	0.00076	0.00064	–	0.00075	0.00076	0.00076
Yeast-cdc	0.00106	0.00092	0.00106	0.00107	0.00107	0.00106
Yeast-diau	0.00419	0.00363	0.00419	0.00414	0.00417	0.00416
Yeast-dtt	0.00340	0.00309	0.00340	0.00348	0.00343	0.00340
Yeast-elu	0.00108	0.00091	0.00109	0.00107	0.00110	0.00109
Yeast-heat	0.00506	0.00441	0.00506	0.00497	0.00507	0.00512
Yeast-spo5	0.02267	0.01879	0.02267	0.02128	0.02163	0.02163
Yeast-spoem	0.02778	0.02564	0.02778	0.02774	0.02879	0.02861
FG-Net	0.08668	0.07519	0.08668	0.08150	0.07551	0.08700
SJAFFE	0.01852	0.02854	0.01852	0.02202	0.01909	0.01881
Music	0.02612	0.02093	0.02612	0.02455	0.02527	0.02384
Painting	0.14665	0.13376	–	0.14330	–	–
SCUT-FBP	0.15590	0.13538	0.15511	0.15752	0.13871	0.15637
fbp5500	0.06766	0.16142	0.06932	0.06405	0.06683	0.06812
emotion6	0.20804	0.18777	0.20785	0.20936	0.21338	0.21285
RAF-ML	0.20525	0.24233	0.20525	0.18498	0.20816	0.20816
Movie	0.04863	0.04554	0.04387	0.04586	–	–
Flickr	0.25063	0.22598	0.25063	0.24809	0.25450	0.25066
Twitter	0.23219	0.20585	0.23220	0.22474	0.22963	0.23639
MOR-PH	0.07911	0.07586	0.07911	0.07123	0.05723	0.05723
Human-Gene	0.01741	0.01473	0.01741	0.01767	0.01756	0.01742

and the variation in predictive changes across repeats. The fifteen cost ties correspond to retaining the full set and therefore reproduce the full-set predictions. Empty outperforms Full on several tasks, distinguishing the predictive utility of the selected attributes from training-reconstruction preservation.

GBFRS-EQ achieves both lower mean cost and lower mean held-out Brier error than GBIP-O1 on sixteen of the 23 datasets. These results do not establish a general cost or predictive advantage for GBIP-O1. The comparison measures prediction on held-out objects, while the O1 constraint applies to training reconstruction; the objectwise checks above quantify the distinct requirement imposed by O1.

Figure 5 illustrates GBIP-O1 cost savings and test Brier changes relative to Full on 23 complete tasks. Panel (a) averages $100[1 - C(P)/C(A)]$ over splits. In (b), circles give split-level $100[B(P) - B(A)]/B(A)$ and diamonds their mean; negative values improve prediction. Markers can overlap. Yeast-alpha and Painting have incomplete results and are excluded.

M2B boundary case. M2B is excluded from the main-task and feature-family comparisons under the declared protocol, rather than selected or excluded after inspecting this table. GBIP-O1 retains all 250 attributes in each split and reproduces Full's predictions. Their mean Brier error is 0.31728, compared with 0.27503 for Empty. This illustrates a limitation of the specified five-neighbor predictor on these splits; it does not show that the attributes are intrinsically uninformative. The three split-level results and all comparator configurations are in `all_results.csv` of the light archive.

7.2.3. Sensitivity to Predictor Feature Scaling

Qian et al. [9] evaluated selected features with AA-kNN at $k = 5$ under ten-fold cross-validation. Our unified predictor uses the same AA-kNN averaging rule [1] and specifies Euclidean distance after training-fold min–max scaling. The original GBFRS experimental description leaves the predictor input scaling unspecified. We freeze the subsets, selector parameters, costs and splits, and apply the same feature transformation to both methods before prediction. The alternatives use the released feature coordinates directly or apply training-fold z-score scaling. At matched min–max scaling, the AA-kNN calculation reproduces the unified predictor to numerical precision.

Table 4 reports the number of tasks on which GBFRS-EQ has lower cost, or both lower cost and lower Brier error, than GBIP-O1 under a shared

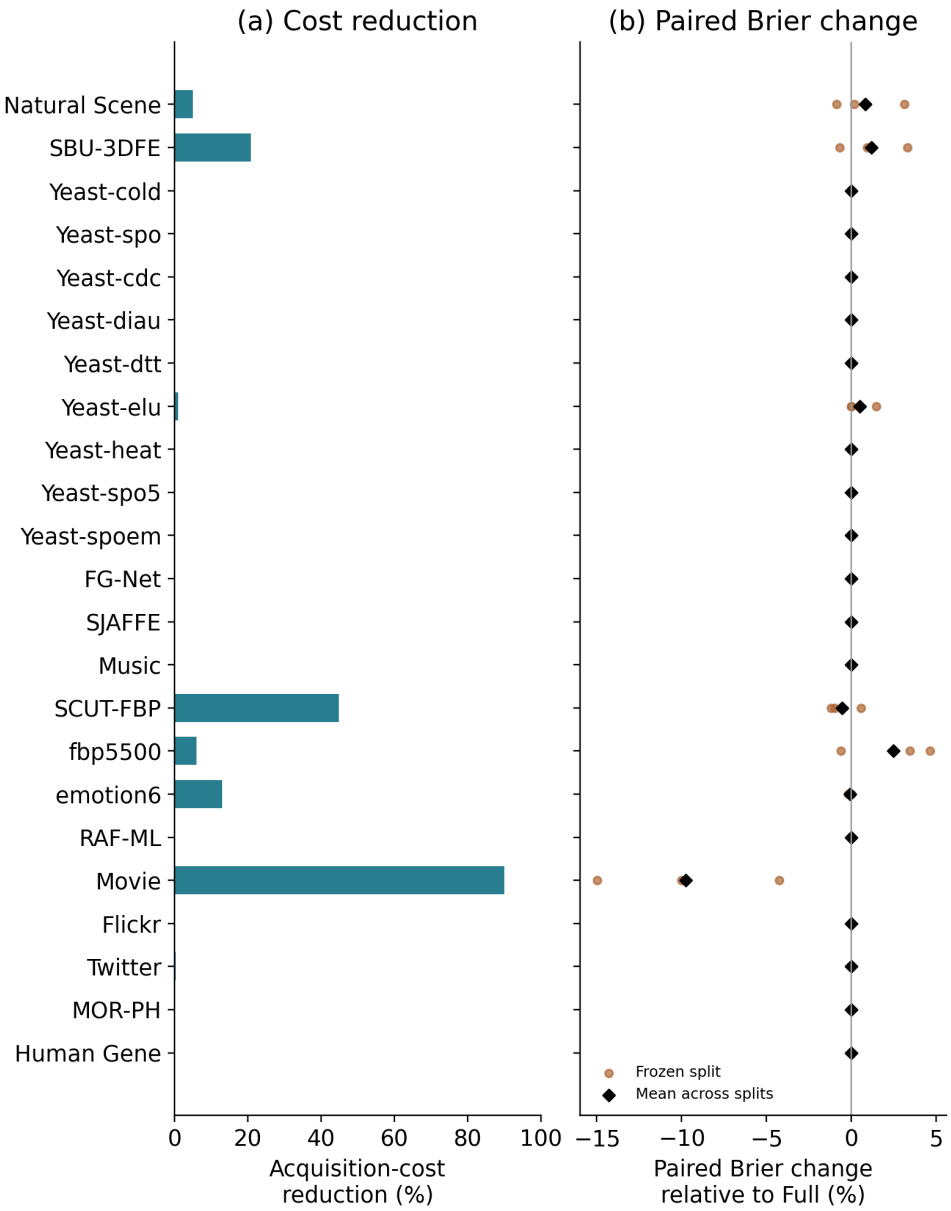


Figure 5: GBIP-O1 outcomes on the 23 tasks with complete three-split means. Left: simulated cost saving relative to Full. Right: relative test-Brier change; negative values improve prediction. Circles are splits and diamonds are their means.

predictor input scaling. Counts use the 23 tasks with three complete paired splits.

Table 4: Predictor Scaling Sensitivity

Input Scaling	Lower Cost	Lower Cost and Brier
Training-fold min–max	22	16
Released coordinates	22	13
Training-fold z-score	22	14

GBFRS-EQ has lower cost and Brier error on a majority of tasks in each scaling setting, with 11 tasks common to all three sets (Table 4). These comparisons isolate scaling at prediction time. They retain the parameters and subsets obtained under the main protocol, so they do not evaluate retuning or reproduce the original ten-fold experiments.

7.2.4. Paired Comparisons across Feature Families

We conduct exploratory paired comparisons at the feature-family level [43]. Each task first averages its three frozen repeats. Within a family, each method then averages the same set of tasks with complete paired results. The nine eligible Yeast conditions contribute one block, and Yeast-alpha and Painting lack complete GBIP-O1 repeats. The GBFRS-EQ comparison thus uses 23 tasks across 15 family blocks. We use two-sided Wilcoxon signed-rank tests [44] with exact sign enumeration. Differences are rounded to 12 decimal places, values of magnitude at most 10^{-12} are treated as zero and omitted from signed ranks, and equal absolute differences receive average ranks. Holm adjustment [45] covers the two primary paired outcomes, normalized cost and Brier error; it does not correct every exploratory analysis in this paper. Normalized cost is expressed as a percentage of the split's full-set cost.

Table 5 reports feature-family comparisons of GBFRS-EQ with GBIP-O1 under the main min–max protocol. Differences are GBFRS-EQ minus GBIP-O1, so negative values favor GBFRS-EQ. Cost differences are in percentage points. L/T/H counts lower, tied and higher outcomes. r_{rb} is the signed rank-biserial effect. An asterisk marks Holm-adjusted $p < 0.05$.

The normalized-cost difference meets the adjusted significance threshold, whereas the Brier difference does not (Table 5). Excluding the partially observed Yeast family gives the same significance decisions. A Friedman

Table 5: Feature-Family Paired Comparisons

Outcome	L/T/H	Median Δ	r_{rb}	Holm p
Normalized cost	14/0/1	−68.04	−0.983	0.000244*
Brier error	10/0/5	−0.00157	−0.533	0.0730

test [46] of Brier error compares GBIP-O1, GBFRS-EQ, Full, IWDLFS and IWCLFS on the common 14 families, after excluding Movie for unavailable IW results. The test gives $Q = 4.384$ and permutation $p = 0.3617$ (49,999 within-family permutations; seed 20261109; $p = (N_{\geq} + 1)/(49999 + 1)$), so the omnibus test does not trigger post-hoc tests. Failure to reject is not evidence of equivalent predictive performance. The tests treat feature families as independent blocks. Their exploratory interpretation is limited to the common coverage of this benchmark.

7.2.5. Mean Preservation versus Objectwise Preservation

A training-only diagnostic changes the feasibility condition while keeping the oracle, full-set reference, costs, deletion order, complete-pass rule, tolerance and candidate-request limit fixed. The alternative accepts P when $n^{-1}\sum_i e_i(P) \leq n^{-1}\sum_i e_i(A) + 10^{-12}$. The two first-repeat cases were chosen from earlier development results to cover full retention (Yeast-cold) and cost savings (SCUT-FBP). These development-selected cases illustrate the effect of changing the constraint. The table includes the original Full references. The objectwise replays reproduce the frozen selections and full-reference error vectors exactly. Actual request counts differ with the deletion paths.

Table 6 reports training reconstruction and attribute cost under mean and objectwise preservation on two first frozen splits. The column "Exceeding" counts objects satisfying $e_i(P) > e_i(A) + 10^{-12}$. Costs are simulated; Full supplies the reference.

The complete-reference mean errors are 0.00599083 and 0.12573394, respectively. In Table 6, mean preservation achieves a lower cost and lower mean reconstruction error, but permits deterioration for 9 and 24 training objects. The maximum increases are 0.00601106 and 0.18196906. GBIP-O1 produces no increases beyond the specified tolerance in these comparisons. Figure 6 shows all training-object changes, including improvements and ties. All nine Yeast-cold objects with increased error fall outside the predefined upper quartile by deviation from the global label mean.

Table 6: Mean versus Objectwise Preservation

Task	Constraint	Cost	Attributes	Mean Training Error	Exceeding
Yeast-cold	Full	127	24	0.00599083	0/192
	Objectwise	127	24	0.00599083	0/192
	Mean	12	3	0.00591508	9/192
SCUT-FBP	Full	1690	300	0.12573394	0/192
	Objectwise	934	172	0.12244419	0/192
	Mean	2	1	0.11635585	24/192

Figure 6 illustrates objectwise changes in training reconstruction error under two preservation rules on two exploratory splits. Each point is an object; $e_i(P) - e_i(A) > 0$ denotes deterioration and a negative value improvement. Horizontal positions follow the frozen training-object order and do not represent a difficulty ranking. The panels use different vertical scales.

7.3. Search and Geometry under Preservation Requirements

7.3.1. Cost Savings under a Shared Objectwise Constraint

We compare GBIP-O1 with GBFRS+O1, the lowest-cost feasible prefix of the GBFRS-EQ ranking under the same training error reference and tolerance. Among the 73 paired splits with valid outputs for both procedures, GBIP-O1 has lower cost on 22, the same cost on 49, and higher cost on two. Both outputs satisfy the objectwise condition in every pair. The two degenerate GBIP-O1 splits are excluded. This comparison holds feasibility fixed and compares two search strategies, while Section 7.2.2 compares costs and predictions against selectors without the shared constraint.

Table 7 reports Movie cost decomposition and deletion-order diagnostic. In panel A, the first two savings are normalized by full-set cost, whereas the third is normalized by the cost of training-nonconstant attributes. In panel B, both endpoints are evaluated using the original prices. Costs and candidate evaluations are three-split means; evaluations exclude the independent terminal-neighborhood checks.

Movie has 1,442–1,457 columns that are constant on the respective training splits. The constant-only baseline drops these columns while keeping the original fitted reference and full-attribute denominator. Its training error vector matches Full exactly, and its test Brier matches Full on all three splits. Constant-column removal accounts for 86.10–86.64% of the cost saved

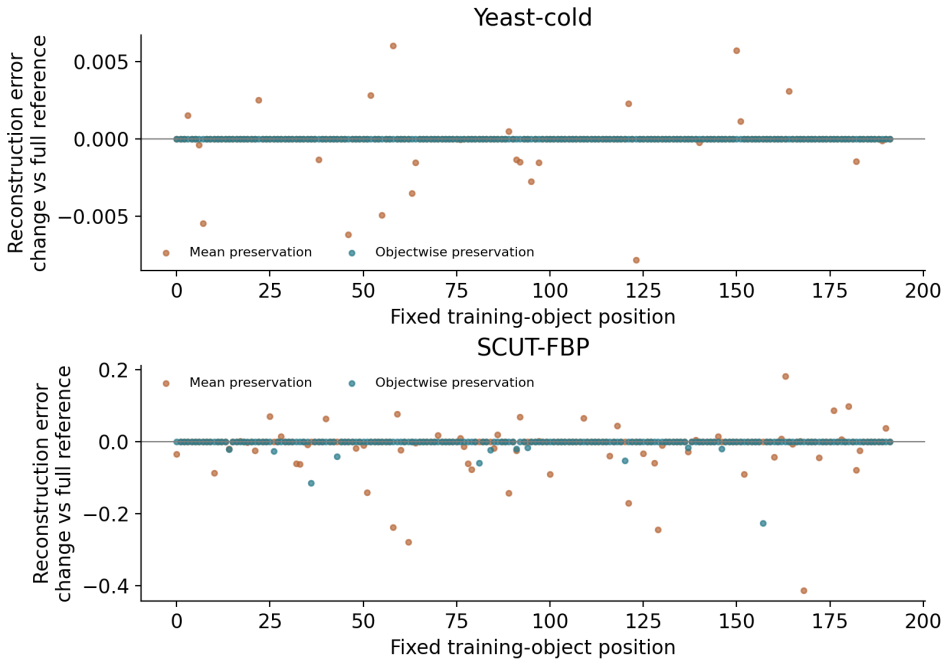


Figure 6: Training-object squared-error changes relative to Full on two first frozen splits. Positive values denote deterioration; panels have different vertical scales. Objects follow frozen training order, not a difficulty ranking.

Table 7: Movie Cost Decomposition and Deletion Order

A. Movie cost savings (%)

Split	Constant-Only Savings	Total GBIP-O1 Savings	Savings Within nonconstant Attributes
1	77.01	89.44	54.06
2	78.13	90.18	55.08
3	78.14	90.74	57.62

B. Deletion order under the same prices

	Cost Retained (%)		Candidate Evaluations	
Task	Descending Cost	Ascending Index	Descending Cost	Ascending Index
Movie	9.88	10.16	2443.3	2514.7
SCUT-FBP	55.09	47.12	1313.7	1327.0

by GBIP-O1, which also removes 54.06–57.62% of the cost of the remaining columns (Table 7A). In each split, 295–337 training-constant columns vary on the test set. The designation is therefore specific to the training sample. On split 1, the extra 12.43 percentage points saved beyond constant removal equal 54.06% of the remaining cost; the percentages use different denominators.

To isolate the order effect on the two largest-saving tasks, we compare descending cost with ascending feature index. The latter is the equal-cost run from Section 7.4.3, where all ordering ties are resolved by index. The two runs share the start, fitted oracle, fixed reference and complete-sweep stopping rule. We evaluate both endpoints at the original prices. All six pairs have verified feasible and single-deletion-irreducible endpoints. Cost ordering gives lower retained cost on all three Movie splits and on one SCUT-FBP split. It uses fewer requests on average on both tasks, while the index order has lower mean retained cost on SCUT-FBP (Table 7B). Descending-cost deletion does not uniformly produce the lower-cost endpoint under nonmonotone feasibility.

7.3.2. Effect of Label-Information Geometry

We compare the full label-information geometry with the unweighted symmetric sum of min–max-normalized per-attribute distances, using the original dimension as denominator. Both variants use the same costs, granular reconstruction, anchored extension and complete deletion sweeps. Each fits its own full-set reference. We also score both selected subsets under the full geometry against its original full-set reference. This joint ablation removes the information weights and their associated directionality. Because each geometry defines its own reconstruction evaluator and reference, the comparison measures their combined effect on reconstruction and reduction, rather than isolating search under an identical feasible set.

On the first frozen splits, the two geometries select identical subsets on Movie and Yeast-cold, at 10.56% and 100% of full cost, respectively. The first SCUT-FBP split shows a cost difference, motivating a follow-up on its two remaining frozen splits with the same protocol. Mean full-set reconstruction error agrees between geometries within each tested split. In this two-geometry comparison, all selected subsets satisfy their own constraints and pass the original full-geometry check. The reverse check fails for the full-geometry subset on SCUT-FBP split 3, where six training objects exceed the raw-geometry reference.

Table 8 reports the effect of label-information geometry on the three frozen SCUT-FBP splits. Retained cost is a percentage of full cost. Differences are unweighted symmetric minus full geometry, in percentage points; positive values favor full geometry. Each fits its own reference; both types of endpoint in this table pass the original full-geometry check.

Table 8: Label-Information Geometry Ablation

	Cost Retained (%)			Test Brier	
Split	Full Geometry	Unweighted Symmetric	Cost Difference	Full Geometry	Unweighted Symmetric
1	55.27	62.84	+7.57	0.16610	0.16611
2	47.87	46.69	−1.18	0.15487	0.15689
3	62.13	59.41	−2.72	0.14436	0.14603
Mean	55.09	56.31	+1.22	0.15511	0.15635

Full geometry lowers retained cost on the first SCUT-FBP split and raises it on both follow-up splits (Table 8). It makes more candidate requests in all three splits, averaging 1,314 versus 1,127, and yields slightly lower test Brier in each split. The ablation changes information weighting and directionality together.

7.3.3. Reciprocal Aggregation and Raw Geometry: Additional Runs

A follow-up on 13 November 2026 reran Yeast-cold, SCUT-FBP and Human-Gene on their three existing frozen splits, with 192 training objects per split. These are exploratory checks on development-consulted datasets, not new independent confirmation. We compared the original maximum of the two directed dissimilarities, their arithmetic mean using the same fitted information weights, and the unweighted raw geometric sum. The original dimension m remains the denominator. Each variant fits its own full-set reference; all costs, splitting parameters and complete-sweep deletion rules are fixed. Replacing the reciprocal maximum by a mean changes the symmetric geometry supplied to the granulator. The subsequent raw-geometry contrast removes weighting; these nested contrasts do not constitute a complete factorial isolation of weighting and directionality.

All 27 selections were frozen before the follow-up test-label scoring. Yeast-cold and Human-Gene retained all attributes under all three geometries on all splits, giving identical predictions within each split. Their mean test

Brier errors were 0.00770576 and 0.01741304, respectively. Table 9 reports SCUT-FBP. These three known splits support descriptive comparisons only.

Table 9: Additional SCUT-FBP Geometry Comparison: Three-Split Means

Reciprocal Aggregation / Geometry	Cost Retained (%)	Test Brier
Original information-weighted maximum	55.09	0.15510788
Information-weighted arithmetic mean	57.69	0.15547805
Unweighted raw geometric sum	56.31	0.15634544

All 27 endpoints satisfied their own objectwise bounds and fresh terminal-neighbor evaluations against the same fitted evaluator. Across 81 endpoint–evaluator combinations, SCUT-FBP split 3 supplied the only cross-reference violations: the arithmetic-mean endpoint violated the original maximum-geometry bound for 19 objects; the original and arithmetic-mean endpoints each violated the raw-geometry bound for six objects. The other combinations had no violations. Thus feasibility must be stated for its specified evaluator and reference, even when two variants have similar aggregate errors.

7.4. Feasibility and Sensitivity

7.4.1. Why Do Some Searches Retain the Full Set?

For the fifteen datasets retaining full cost in all three repeats, all 4254 retained-attribute deletion checks violate the objectwise condition. Each retained attribute was tested separately at the endpoint. The number of deteriorating objects in an individual check ranges from 1 to 44 across these records.

Because the constraint is nonmonotone, single-deletion blocking does not exclude feasible simultaneous deletions. Exhausting all 276 attribute pairs on Yeast-cold repeat 1 finds no feasible pair. A further fixed-budget diagnostic covers the other seven full-retention feature families: FG-Net, SJAFFE, Music, RAF-ML, Flickr, MOR-PH and Human-Gene. Each uses all three frozen splits and 64 unique subsets per split, fixed before oracle evaluation: 16 double deletions, 16 triple deletions, 16 subsets retaining $\lfloor m/2 \rfloor$ attributes and 16 retaining $\lfloor m/4 \rfloor$. All 21 full-reference vectors match exactly. None of the 1344 candidates is feasible; the minimum number of harmed objects over the 192 candidates per family is 3, 3, 4, 6, 9, 4 and 18, respectively.

The single-deletion checks certify local irreducibility. The joint-deletion findings apply to the specified candidates, with Yeast-cold as the only Yeast condition tested.

7.4.2. Sensitivity to Training-Sample Size

We extend three feature families using nested training samples while retaining each repeat's original 256 test objects. The original 192 training objects form the prefix; a predetermined permutation adds objects outside that training/test split. Yeast-cold uses $n = 192, 384, 768, 2209$, SCUT-FBP uses $n = 192, 384, 768, 1244$, and Human-Gene uses $n = 192, 384, 768$. The largest Yeast-cold and SCUT-FBP samples contain every object outside the fixed test set; Human-Gene remains capped at 768. All attributes, acquisition costs, representation parameters and deletion rules remain unchanged. Each size refits its geometry and full-set reference. Consequently, reference vectors differ across sizes rather than defining a common cross-size error threshold.

All 33 size–repeat cells complete, producing 99 Full, GBIP-O1 and mean-preservation evaluations. The nine 192-object references and GBIP-O1 selections reproduce the previous outputs exactly. An independent recomputation from the frozen subsets reproduces the four reported prediction metrics within 10^{-12}. The supplement documents the selection hashes and reconstruction checks.

Table 10 reports sensitivity to nested training size across three frozen repeats. Costs are percentages of full acquisition cost; GBIP-O1 reports mean $\pm$ sample standard deviation. For the Mean selector, the "Exceeding" column gives the mean percentage of objects whose errors exceed their respective references by more than 10^{-12}. Each size fits its own reference.

Table 10 shows that acquisition savings are not stable under this sample expansion. On SCUT-FBP, mean cost retained rises from 55.09% at 192 objects to 97.36% at 384, then 100% in all three repeats at both 768 and 1244. Yeast-cold and Human-Gene retain the full cost at every tested size. All nine GBIP-O1 outputs at the largest size of each family are therefore full-set outputs and reproduce Full's held-out predictions. The small SCUT-FBP predictive improvements at 192 and 384 disappear when the output becomes the full set (Figure 7). All 33 GBIP-O1 outputs satisfy the recorded objectwise condition, but this feasibility cannot be interpreted as effective reduction.

Figure 7 presents the nested training-size diagnostic. Lines connect three-repeat means; faint points show repeats. The upper panels report retained

Table 10: Training-Size Sensitivity

Task	n	O1 Cost (%)	Mean Cost (%)	Mean-Selector exceeding (%)
Yeast-cold	192	100.00 ± 0.00	7.09	7.47
Yeast-cold	384	100.00 ± 0.00	6.82	8.77
Yeast-cold	768	100.00 ± 0.00	5.77	7.99
Yeast-cold	2209	100.00 ± 0.00	32.55	9.37
SCUT-FBP	192	55.09 ± 7.13	0.36	12.67
SCUT-FBP	384	97.36 ± 2.35	5.70	13.72
SCUT-FBP	768	100.00 ± 0.00	0.85	14.28
SCUT-FBP	1244	100.00 ± 0.00	1.14	13.88
Human-Gene	192	100.00 ± 0.00	1.89	18.58
Human-Gene	384	100.00 ± 0.00	3.78	21.01
Human-Gene	768	100.00 ± 0.00	59.97	12.07

cost; the lower panels report the test Brier difference $B(P) - B(A)$ relative to Full fitted on the same sample. Negative differences indicate lower Brier error. Test objects remain fixed; each size refits its geometry and reference. Human-Gene remains capped at 768.

Mean preservation often admits substantially lower-cost subsets while allowing individual deterioration (Table 10). It is not uniformly predictable from sample size: for example, its Human-Gene cost at 768 varies strongly across repeats, as the individual points in Figure 7 show. Larger samples change the objectwise requirements, fitted geometry and reference representation together; the present design does not isolate their causal contributions. The results show that the observed cost savings can depend on training-sample size.

7.4.3. Sensitivity to Simulated Attribute Costs

We distinguish repricing a fixed subset from rerunning the cost-ordered search. Both analyses use the original data splits and express subset cost as a fraction of the full cost under the same vector. For repricing, we apply the original costs, equal costs, two draws of uniform integers from 1 to 10, and two lognormal draws with log-scale standard deviation 1.5. The first panel of Table 11 uses the 23 tasks with three valid GBIP-O1 outputs. For fixed subsets and positive prices, the same eight tasks necessarily retain savings; the sensitivity concerns their magnitude. GBFRS-EQ is cheaper and has

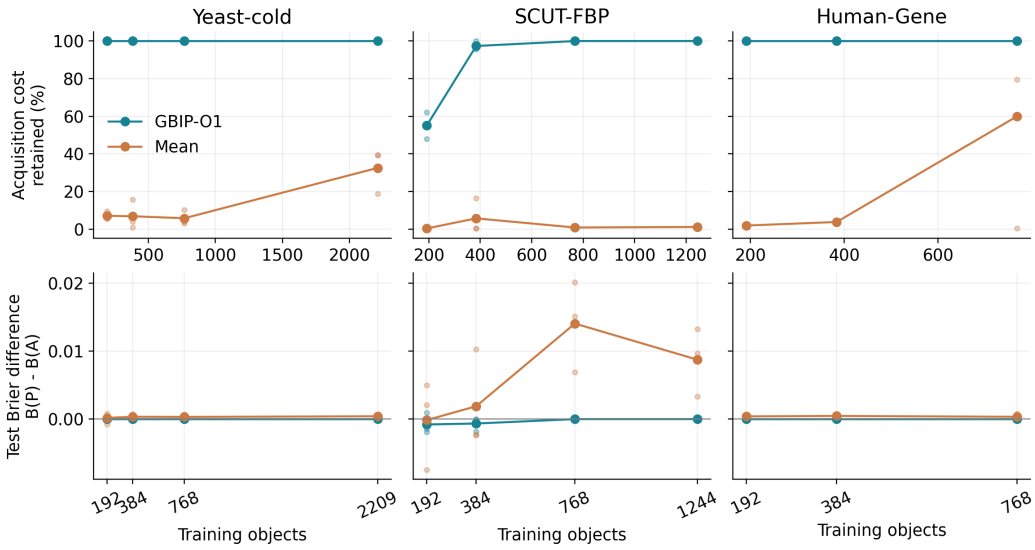


Figure 7: Nested training-size sensitivity with fixed test objects. Lines are three-split means and faint points are splits. Top: retained simulated cost. Bottom: test-Brier difference from Full fitted on the same sample; negative is better. Each size refits its geometry and reference.

lower Brier error on the same sixteen tasks under all six settings. Movie and SCUT-FBP account for 72.10–77.85% of total savings across these settings.

Table 11 reports the effects of attribute-cost assumptions on fixed-subset valuation and cost-ordered search. Costs are normalized by full-set cost under the corresponding prices. Panel A reprices fixed subsets and averages over 23 tasks or 15 families. Panel B reports three-split rerun means; parenthesized values give the original subset's cost under the corresponding rerun prices. Weight-aligned/reversed costs are simulated stress tests.

We rerun GBIP-O1 on all three original splits of Movie and SCUT-FBP, the two tasks with the largest original savings. The seven alternative vectors add two stress tests to the equal and random costs. These assign the same lognormal prices in increasing or decreasing order of $n^{-1}\sum_i W_a(o_i)$, with original attribute index breaking equal-weight ties. The weight alignment is a simulated proxy for systematic cost variation. All 42 alternative searches change the selected subset and end at feasible, single-deletion-irreducible outputs. Under matched prices, 23 are cheaper than the original subset, 18 are more expensive, and one ties. The search panel shows selected settings. In Movie, the aligned and reversed vectors yield substantially different retained

Table 11: Attribute-Cost Sensitivity

Fixed subsets: GBIP-O1 cost saving (%)		
Cost Setting	Task Mean	Family Mean
Original	7.88	12.02
Equal	7.70	11.73
Uniform draws 1 and 2	7.64–7.65	11.67
Lognormal draws 1 and 2	7.01–8.01	10.70–12.22
Reordered search: GBIP-O1 cost retained (%)		
Cost setting	Movie	SCUT-FBP
Original	9.88 (9.88)	55.09 (55.09)
Equal	10.01 (9.95)	46.11 (56.56)
Weight-aligned	48.16 (48.73)	65.39 (66.10)
Weight-reversed	0.47 (0.47)	48.98 (60.37)

costs even when the original subset is held fixed. Across the benchmark, 51 splits reject every single deletion in the original first pass. Because price only changes the scan order, the single-deletion procedure returns the full set on these splits under every positive price vector, with the oracle, tolerance and reference fixed. Global exclusion of proper subsets would follow under the additional assumption that each object's error is nonincreasing as features are added: every proper subset lies inside at least one rejected single deletion and would inherit that deletion's violating object. Regenerating the reconstruction for each subset leaves that monotonicity assumption unestablished.

The supplementary cost-sensitivity records contain the split-level results and budget–preservation curves; they are distinct from the six self-contained reviewer checks. For each budget, a curve shows the smallest observed worst-object error increase over verified candidates within that budget. The curves describe the queried candidate pool. The two-task reruns and the limited number of random draws delimit the scope of the sensitivity comparison.

7.4.4. Sensitivity to Objectwise Tolerance

We vary the additive tolerance $\tau \in \{0, 0.001, 0.01, 0.05, 0.1\}$ in the squared training reconstruction loss, with 10^{-12} numerical slack. Movie, SCUT-FBP and Yeast-cold each use their first frozen split. Each run starts from the full set with the original costs, geometry and reference. The strict setting reproduces the original search, and every endpoint passes its own

tolerance and terminal single-deletion checks. Table 12 contrasts the strict setting with $\tau = 0.01$. Full grid results, worst-object increases and the sensitivity plot appear in the supplement.

Table 12 reports the effect of reconstruction tolerance on GBIP-O1 outcomes on the first frozen split of three tasks. Each tolerance starts a new full-set search under the same geometry, reference and prices. Costs are percentages of full cost. The column "Exceeding" counts objects with error increases beyond numerical slack at $\tau = 0.01$, not violations of that relaxed bound.

Table 12: Reconstruction-Tolerance Sensitivity

	Cost Retained (%)			Test Brier	
Task	Strict	$\tau = 0.01$	Exceeding	Strict	$\tau = 0.01$
Movie	10.56	7.41	9/192	0.04552	0.04782
SCUT-FBP	55.27	56.45	4/192	0.16610	0.16203
Yeast-cold	100.00	0.00	15/192	0.00721	0.00644

On SCUT-FBP, the larger feasible set leads to a higher-cost endpoint at $\tau = 0.01$, because the greedy deletion path changes. Yeast-cold retains all attributes at $\tau = 0.001$ and returns the empty set at $\tau = 0.01$, with a maximum training error increase of 0.00601. These are experimental tolerances for the stated loss. An application threshold would require a task-specific justification.

7.5. Decision Consequences

7.5.1. Reconstruction Error and Dominant-Label Changes

We reconstruct the frozen Full, GBIP-O1 and mean-preservation outputs for all nine 192-object cases in the three sample-size families, with their eighteen 384/768-object cases as supplementary diagnostics. This fixed 27-case analysis performs no new subset selection and reproduces the full error vectors and three methods' mean reconstruction errors within 10^{-12}. Counts refer to training-object occurrences; an original object can recur across splits.

A dominant-label transition is assessed only when the target and reference have unique maxima at tolerance 10^{-12}. A change from the correct reference maximum to a unique incorrect maximum is counted separately from a predicted tie. For each object, pairwise ordering loss averages over label pairs (a, b) with $\ell_i^a > \ell_i^b$: a predicted reversal contributes one, a predicted

tie contributes one half, and a correct order contributes zero. Target ties are excluded. Ties use the stated 10^{-12} tolerance; comparisons count object-level loss increases rather than pooling all label pairs across objects. These measures describe the reconstructed training vectors; the primary evaluation remains the distributional error specified by the constraint.

Table 13 reports dominant-label transitions across the three 192-object repeats of each family (576 object occurrences per row). $C \to W$ denotes a correct unique reference maximum becoming uniquely incorrect; $W \to C$ denotes the reverse. No correct-to-tied transitions occur in these cases.

Table 13: Dominant-Label Transitions

	Mean Error	Mean Selector		GBIP-O1	
Family	increases	$C \to W$	$W \to C$	$C \to W$	$W \to C$
Yeast-cold	43	21	35	0	0
SCUT-FBP	73	24	28	2	13
Human-Gene	107	37	42	0	0

Among the 223 mean-selector error increases in Table 13, 69 include a correct-to-incorrect dominant-label transition, while GBIP-O1 keeps the correct maximum in those 69 occurrences. Their median error increases are 0.00182845, 0.04151158 and 0.00047333 in the three families, respectively. All 69 increases are at least 10^{-6}. The mean-selector improvements elsewhere are also retained: 105 uniquely incorrect reference maxima become correct in the nine cases. The cost premiums for GBIP-O1 over the mean selector average 92.91, 54.73 and 98.11 percentage points of full subset cost.

GBIP-O1 itself yields two correct-to-incorrect transitions on SCUT-FBP repeat 1. At source rows 659 and 470 (zero-based indices), the target and Full maxima are at component 2 (zero-based indexing) and the GBIP-O1 maximum is at component 1. Nevertheless, squared errors decrease from 0.03147797 to 0.00513887 and from 0.02681575 to 0.00406137. Thus the objectwise squared-error certificate can hold while a correct maximum changes. Pairwise-ordering deterioration also occurs: GBIP-O1 has three such occurrences at 192 objects and two at 384 objects on SCUT-FBP. Full object-level records, reverse transitions and supplementary sizes accompany the results.

The squared-error bound can hold even when an object's dominant label or label ordering changes. Applications that require either decision to be preserved need a corresponding acceptance criterion.

Decision Stability Among O1-Feasible Occurrences. Table 14 separates empirical decision stability from the guarantee supplied by the loss bound. For Mean, the denominators are restricted to its individually O1-feasible occurrences. These conditional summaries characterize each selector's outputs on its own feasible set, rather than a matched-object comparison between columns.

Table 14 reports decision diagnostics for nine 192-object cases. Counts are training-object occurrences; objects may recur across splits. Apart from the first row, denominators use each method's O1-feasible occurrences. The unique-correct row additionally requires a unique correct reference maximum. Certified counts use feasible occurrences, not only unique-target objects. Pairwise loss uses the stated tie rule.

Table 14: Decision Diagnostics and Certificate Coverage

Diagnostic	GBIP-O1	Mean
Meets the O1 bound	1728/1728	1505/1728
Preserves the reference maximum set	1712/1728	1305/1505
Preserves a unique correct maximum	643/645	494/507
No increase in pairwise ordering loss	1725/1728	1493/1505
Unique correct maximum certified by the margin	261/1728	188/1505

Margin-certificate coverage. Proposition 8 certifies a unique correct maximum for 261 of the 1,728 GBIP-O1 occurrences in the nine 192-object cases. All 261 retain that maximum. The remaining occurrences lie outside this sufficient certificate and may still preserve their decisions empirically. The certificate uses observed training targets and does not certify unlabeled test objects.

7.6. Image-Pipeline Case Study

7.6.1. Objectwise Preservation and Storage on SCUT-FBP5500

The SCUT-FBP5500 benchmark provides per-image rating distributions for facial attractiveness prediction [5]. We use the official 3,300/2,200 train/test split and form each five-bin target from all 60 ratings. A frozen ImageNet-pretrained ResNet-18 [47] supplies 512 float32 coordinates per image. The recorded extractor uses torchvision's `ResNet18_Weights.IMAGENET1K_V1`, its associated `weights.transforms()`

Table 15: SCUT-FBP5500 Reconstruction and Resource Use

Method	Attributes Retained	Exceeding	Mean Train Error	Test Brier	Test Rating MAE	MB	Backend Time (s)
Full	512	0	0.12201	0.10645	0.38116	6.7584	2.3704
Empty	0	859	0.15467	0.15055	0.55973	0	0.0042
GBIP-O1	512	0	0.12201	0.10645	0.38116	6.7584	2.3504
Mean	51	726	0.12108	0.12671	0.44148	0.6732	0.4388

preprocessing and an identity replacement for the final fully connected layer; the benchmark checks the weight-file hash against its extractor manifest. This setup is separate from the `fbp5500` array in the broad benchmark. Full, Empty, GBIP-O1 and Mean share the same frozen inputs, with no tuning on the official split. GBIP-O1 and Mean use the constraints in Sections 6 and 7.2.5, respectively. Held-out predictions use the same five-nearest-neighbor rule with training-fitted min–max scaling, with the training mean distribution used for Empty. Each retained coordinate requires four bytes per stored training image. Rating MAE is the mean absolute difference between target and predicted expected scores, using rating levels 1–5.

Table 15 reports training reconstruction, test prediction and descriptor storage on the official SCUT-FBP5500 split. The column "Exceeding" counts objects whose reconstruction errors exceed their reference errors by more than 10^{-12}. Storage is $4 \cdot 3300 \cdot |P|$ bytes, in decimal MB, excluding images, model, labels and auxiliary data. Backend time is the median for all 2,200 test objects.

Table 15 shows that Mean slightly improves the average training reconstruction error while increasing it for 726 of 3,300 objects (22.0%). Errors decrease for 713 objects, and the largest increase is 0.56036. The 51-coordinate model reduces raw feature storage by 90.04% and measured backend time by 81.49% relative to Full, with increases in held-out Brier error and rating MAE. Timing uses seven interleaved measurements per method after two warm-ups and covers normalization, coordinate selection, distances, sorting and aggregation. Image decoding and feature extraction are outside this timing scope.

GBIP-O1 retains all 512 coordinates, yielding the same training errors, predictions and storage as Full. Their small timing difference reflects measurement variation rather than different feature computations. Replaying the selection predicate on the saved error vectors confirms that each of the 512

single deletions violates the bound for 214 to 733 objects. We also check the 1,017 distinct nonempty subsets formed by joint deletions and queried along the Mean search. Each violates the bound for at least 658 objects. The replay verifies saved-vector digests and both methods' selection decisions. These joint candidates follow the Mean query path.

An independent follow-up fixes 512 additional candidate subsets before evaluation, using random seed 20261110 and excluding every previously queried subset. The candidate pool contains 256 double deletions, 64 triple deletions, 64 quadruple deletions and 32 deletions of each size 128, 256, 384 and 448. The geometry, reference and 10^{-12} numerical tolerance remain fixed. The full-reference replay matches exactly. None of the 512 candidates is feasible, either at numerical tolerance or at the recorded zero threshold (Table 16). The smallest number of violating objects is 415/3300. The smallest maximum error increase across candidates is 0.50206; its candidate violates the bound for 450 objects. The minimum violating count and minimum maximum-increase summaries need not identify the same candidate.

Table 16 reports feasibility of 512 independently specified joint-deletion candidates on SCUT-FBP5500. Candidates were fixed before evaluation and exclude previous queries. Violating objects exceed their frozen reference by 10^{-12}; min–max gives object-count ranges within each deletion-size group, not error magnitudes.

Table 16: Independent Joint-Deletion Feasibility

Attributes Deleted	Attributes Retained	Candidates	O1-Feasible Candidates	Violating Objects min–Max
2	510	256	0	415–731
3	509	64	0	517–738
4	508	64	0	563–736
128	384	32	0	695–746
256	256	32	0	694–743
384	128	32	0	694–739
448	64	32	0	708–750

Together, the two query pools cover 2,041 distinct nonempty proper subsets, all infeasible under the fixed bound. Subsets outside these pools remain unresolved. The follow-up audit checks the returned summaries against the

frozen protocol and original query manifest; the candidate error vectors remain in the run's SQLite cache.

A follow-up measures the complete query pipeline from the original test images to the predicted distributions (Table 17). It uses CPU inference with four PyTorch threads, batches of 32 and the same frozen training features. Full and Mean are interleaved for two warm-up rounds and seven measured rounds each. Reading the ZIP and decoding images are included, followed by preprocessing, ResNet-18 inference and the original prediction backend. Model loading precedes measurement; repeated passes use warmed file-system caches. The recorded environment is Windows 11 (build 26200), CPU identifier Intel64 Family 6 Model 183 Stepping 1, PyTorch 2.5.1+cpu and torchvision 0.20.1+cpu. The return does not record a marketed CPU model or RAM capacity. Across the seven rounds, total times range from 32.56580 to 35.96900 seconds for Full and 30.73749 to 36.83712 seconds for Mean. The total median falls from 32.90014 to 30.97551 seconds, a 5.85% reduction. The backend median falls by 79.35%, while the shared image pipeline dominates the total. Re-extracted test features differ from their frozen values by at most 5.25×10^{-6}, and predictions match the frozen predictions exactly in every recorded round.

Table 17 reports end-to-end CPU query time for 2,200 SCUT-FBP5500 test images, in seconds. Entries are separate medians of seven interleaved rounds after two warm-ups; four PyTorch threads and batches of 32 are used. ZIP reading through prediction is included; model loading is excluded and file caches are warm. Stage medians need not sum to the total, which includes concatenation and other overhead.

Table 17: End-To-End CPU Query Time

Stage	Full	Mean
Image reading and decoding	1.79564	1.79022
Preprocessing and ResNet-18	28.81379	28.70898
Prediction backend	2.30858	0.47674
Total	32.90014	30.97551

On this split, the storage reduction comes from accepting average-error preservation; O1 retains the full feature matrix. The two backend reductions (81.49% in the isolated backend test and 79.35% within the end-to-end rounds) describe separate timing runs. Selection-stage peak memory and

comparable selection runtimes are not available across all methods, so these measurements do not establish a selection-efficiency advantage. As a storage-order illustration only, caching 512×3300^2 scalars would require about 22.30 GB in float32 or 44.61 GB in float64 before other arrays; this is not a measured peak and does not specify every implementation's cache strategy.

Recorded-Pool Storage Envelope on SCUT-FBP5500. We pool the 1,531 original queries and 512 independent candidates from Section 7.6.1, including Full and Empty. For each of these 2,043 subsets, define $d(P) = \max\{0, \max_i[e_i(P) - e_i^\star]\}$. With $S(P) = 4 \cdot 3300 \cdot |P|$ bytes, the recorded-pool envelope is

$$S_{\mathcal{C}}(\tau) = \min_{P \in \mathcal{C}:\, d(P) \le \tau} S(P). \tag{7.1}$$

Numerical comparisons retain the original 10^{-12} slack. Candidate ties use the lexicographic order of sorted feature indices. The fixed pool yields a nonincreasing storage envelope; the separate greedy reruns above can change paths as tolerance changes. The six endpoints are selected using training reconstruction and storage. The returned endpoint-assessment records report reproduction of all six training-error-vector hashes before decision and held-out assessment. The self-contained public audit checks those saved summaries and flags; it does not independently replay the underlying error vectors.

Table 18 reports minimum-storage endpoints within the recorded candidate pool and their decision and prediction outcomes. L_1 counts losses among 2,159 reference-correct unique top-1 decisions; L_2 among 1,973 eligible reference-correct top-2 sets. R counts ordering-loss increases among 3,300 objects. The "Certified" column applies the reference-plus-tolerance bound. Tolerance knots are displayed to five decimal places for readability; selection and certificate calculations use the unrounded stored values. A displayed knot rounded downward must not be used as an exact feasibility threshold. Full-precision values are in `supplementary/tolerance_envelope/results/frontier_summary.csv` of the reviewer archive.

The decision denominators are 2,159 training objects with a unique correct reference maximum, 1,973 with a correct top-2 target set and unambiguous target and reference cutoffs, and 3,300 with at least one unequal target-label pair. A predicted tie counts as loss of an unambiguous decision. The certificate column uses Eq. (6.6) with the endpoint's allowed tolerance. Top-2 outcomes are assessed directly from the reconstructed distributions.

Table 18: Recorded-Pool Storage Endpoints

τ Knot	Dimensions	MB	L_1	L_2	R	Certified	Test Brier
0.00000	512	6.7584	0	0	0	1479	0.10645
0.48122	511	6.7452	75	92	214	0	0.10635
0.50206	510	6.7320	123	139	392	0	0.10621
0.52498	89	1.1748	184	204	571	0	0.11582
0.53036	50	0.6600	192	224	583	0	0.12720
0.56036	0	0.0000	237	242	760	0	0.15055

No recorded subset reduces storage at $\tau \leq 0.48$. The first reduction retains 511 coordinates at $d(P) \simeq 0.48122$, saving 0.20%. Storage falls to 89 coordinates at approximately 0.52498 and 50 at 0.53036. The latter saves 90.23% of stored-feature bytes, with 192 losses of a reference-correct top-1 decision, 224 losses of a reference-correct top-2 set, and 583 objects with increased ordering loss. At this endpoint, 197 initially incorrect unique reference maxima become correct and 557 objects have reduced ordering loss. The 50-coordinate endpoint differs from the 51-coordinate Mean endpoint measured in Table 17. The empty subset becomes admissible at 0.56036. These rounded descriptions refer to full-precision thresholds in the recorded pool; they are not global compression thresholds. Application tolerances require separate justification.

Figure 8 illustrates storage–reconstruction trade-offs within a frozen pool of 2,043 subsets, including Full and Empty. Panels (a)–(c) assess the lowest-storage recorded feasible subset at each tolerance. Panel (b) uses metric-specific eligible denominators. Panel (d) shows sufficient-certificate coverage under the permitted error bound, not observed accuracy; it may change with the same selected subset. Test outcomes do not select endpoints.

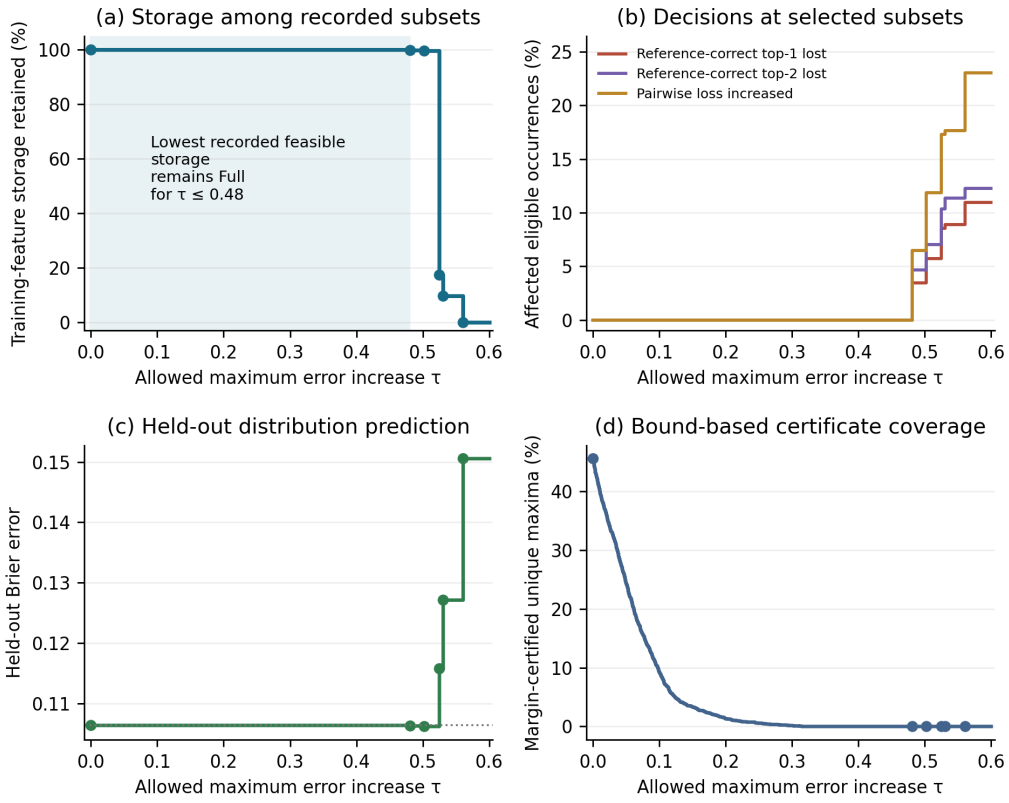


Figure 8: SCUT-FBP5500 recorded-pool storage and reconstruction trade-offs over 2,043 queried subsets. Endpoints are selected using training errors and storage only. Decision curves use metric-specific eligible denominators; certificate coverage uses the allowed bound rather than observed accuracy.

7.7. Additional Original-Data Replay and Measured Selection Resources

The follow-up restored the original MAT files from the frozen upstream revision. Both their byte hashes and the converted-array hashes matched the recorded manifests. For the nine original-geometry runs in Section 7.3.3, selected subsets, full-reference error vectors, final error vectors and test Brier values matched the earlier records exactly. This verifies execution on the available original inputs for these cases; it does not independently validate every baseline or all 25 tasks.

The 27 runs recorded 10,979 distinct queried subsets when counted separately within each run, comprising 2,107,968 actual float64 object-error values. Database integrity and every stored vector hash were checked. These are computation records, not independent observations. The public supplement contains source, protocols, split definitions, hashes and aggregate results at `experiments/additional_20261113/` in the companion repository. Original data, extracted features and complete individual-error records are retained in a private cache with hash manifests; their public redistribution is not implied.

Table 19 reports the original method's three-run means on Linux with an AMD EPYC 9V74 processor, three-core process affinity and an 8 GiB memory limit. Each selection used a fresh Python process, sequential jobs and one BLAS thread (Python 3.12.14, NumPy 2.3.5). CPU time covers geometry fitting and deletion search. Peak resident-set size (RSS) covers the process through the end of selection, including imports, resident training inputs and the error-recording buffer; later independent endpoint checks and SQLite export are excluded. These measurements are specific to $n = 192$ and are not full-image or cross-method memory benchmarks. All three geometry variants used the same fitted cache layout, so raw-geometry RSS does not measure an optimized raw-only implementation.

Table 19: Measured Original-Method Selection Resources at $n = 192$

Feature Family	Fit + Search CPU (s)	Mean Peak RSS (MB)
Yeast-cold	1.716	45.50
SCUT-FBP	101.900	219.05
Human-Gene	2.366	52.98

7.7.1. Official Image-to-Prediction Replay

A separate replay used the official SCUT-FBP5500 image archive, all 60 final ratings per image and the official 3,300/2,200 training/test split. Five-bin label distributions were reconstructed from the final-rating column, with 60 distinct raters verified for every image. All 5,500 images were processed anew with ImageNet-pretrained ResNet-18, using the fixed torchvision preprocessing and an identity final layer. The archive, model, features, labels, split and predictions have recorded SHA-256 identities in the supplement.

We evaluated the six pre-existing recorded-pool endpoints in Table 18, retaining 512, 511, 510, 89, 50 and zero coordinates. Their test Brier, total variation, KL divergence and Jensen–Shannon divergence values all matched the earlier endpoint assessments exactly; in endpoint order, Brier was 0.10644553, 0.10635330, 0.10620608, 0.11581688, 0.12719849 and 0.15054966. Endpoints were fixed before this replay, and test results did not choose among them. The image-byte hashes were unique and the train/test image-hash intersection was empty. Byte uniqueness does not establish subject-disjointness. The official source spreadsheet has an owner-ID field for only 99 images and does not document it as a complete depicted-person identifier; rater IDs identify rating volunteers, not image subjects.

The frozen-endpoint replay covers image extraction and held-out prediction. The separate full-selection rerun below provides additional training-search evidence; neither rerun reproduces the earlier Windows timing experiment or the full relaxed-tolerance query pool.

7.7.2. Fresh Full-Image Objectwise-Preserving Selection

We additionally ran the complete GBIP-O1 deletion search on all 3,300 official training objects and 512 freshly extracted ResNet-18 coordinates. All coordinate costs were one, measuring retained descriptor storage rather than image-extraction cost. The search started from Full, used its own fixed reference and retained the default anchored construction, complete cost/index-ordered sweeps and 10^{-12} numerical slack. Previously queried subsets did not warm-start this search. The official test outcomes were already known from earlier evaluation, so the rerun is reproducibility evidence on a known split, not untouched-test confirmation.

To fit the available 8 GiB memory limit, the implementation caches normalized attributes and local information weights, and streams one float64 subset-distance matrix. A compiled numerical kernel retains the original per-attribute accumulation order, with fast-math and fused multiply–add

disabled. The seed-pair routine finds the exact largest-distance cutoff and sorts its boundary ties with the original distance/index rule, returning the same first 256 pairs instead of sorting every pair. No reconstruction gate or tolerance changes. The full 3,300-object distance matrix and reference error vector match the original Python calculation exactly. Checks also cover multiple small-subset inputs and 18 tie-order cases. After selection, two predeclared single-coordinate-deletion candidates were re-evaluated with the original complete pair-sorting implementation; their actual error vectors matched exactly.

The completed search retained all 512 coordinates. Each of the 512 single-coordinate deletions violated the fixed reference bound, affecting between 214 and 733 training objects. This supplies a local single-deletion-irreducibility certificate at Full; it does not rule out feasible joint deletions or prove a global minimum. The search recorded 513 distinct queried subsets and 1,692,900 actual float64 object-error values. Database integrity, all vector hashes and every final-neighbor verdict were checked. The final test prediction equals Full exactly, with Brier 0.10644553.

Table 20: Full-Image Selection after Reusing the Fitted Geometry Cache

Measure	Observed Value
Training objects / input coordinates	3,300 / 512
Retained coordinates	512
Recorded subset-error vectors	513
Selection wall time (s)	2769.57
Selection process CPU time (s)	3841.51
Selection-process peak RSS (MB)	564.88

The selection process ran on the same Linux/AMD EPYC host, with three-core affinity, a two-CPU cgroup quota, three OpenMP distance threads and one BLAS thread. Table 20 excludes previously cached image extraction and geometry fitting; it includes full-reference evaluation, deletion search, resident selection inputs, the active distance matrix and recording overhead. It is not an end-to-end training peak or a cross-method timing benchmark. Shared-host wall times should not be interpreted as stable speed advantages. The independently recorded image-pool tolerance analysis is unchanged: this rerun covers strict default O1, not reconstruction of the older entire relaxed query pool. Source, frozen protocol and aggregate query records are in

`experiments/official_image_selection_20261113/`; original inputs and individual vectors remain in the private cache.

8. Discussion

The framework separates aggregate performance from individual reconstruction feasibility. GBFRS-EQ has lower acquisition cost and lower test Brier error on most tasks with complete results, whereas matched-reference comparisons isolate the costs of alternative feasible-subset searches (Sections 7.2.2 and 7.3.1). Family-level tests support a cost difference but not a Brier difference at the adjusted threshold (Section 7.2.4). These results support evaluating the specified requirement, rather than a general predictive advantage.

Full retention is common. The fresh 3,300-object image run also retains all 512 coordinates under strict O1 (Section 7.7.2). Single-deletion checks establish local irreducibility, while joint diagnostics cover only the queried subsets. Movie's savings largely remove training-constant columns; the larger-sample experiments eliminate SCUT-FBP's initial savings. Prices affect both subset valuation and deletion order (Section 7.4.3). Thus savings depend on the training representation and acquisition assumptions. Additional reciprocal-aggregation runs also show that feasibility does not transfer automatically between geometric evaluators (Section 7.3.3).

A larger tolerance expands the feasible set without guaranteeing a lower-cost greedy endpoint (Section 7.4.4). Geometry likewise changes costs in either direction on SCUT-FBP, despite slightly lower test Brier under full geometry (Section 7.3.2). In the image case, substantial recorded-pool compression requires appreciable worst-object deterioration and changes training decisions in both directions (Table 18). A reconstruction bound alone does not preserve label decisions [1, 4, 8]; the margin certificate and direct diagnostics distinguish these outcomes.

Resource measures also differ: Mean's 90.04% descriptor-storage reduction yields only a 5.85% end-to-end query-time reduction because feature extraction remains unchanged (Section 7.6.1). The broad benchmark uses simulated attribute costs, development-consulted splits and equation-based baseline reconstructions. Object-level splitting does not establish subject-disjoint generalization. The additional replay verifies source-array hashes for three families and official image inputs, but does not resolve row-to-subject

identity or the provenance of every benchmark array. Independent evaluation with measured costs and original baseline protocols would extend the evidence.

Practical use requires a task-appropriate reference, loss and tolerance. The experiments quantify specified reconstruction requirements without establishing an application acceptance threshold. Held-out prediction and full-pipeline timing provide complementary measures of their consequences.

9. Conclusion

This study formulates cost-sensitive attribute reduction for label distribution learning under fixed-reference objectwise reconstruction bounds. GBIP-O1 combines label-information geometry and granular-ball reconstruction with complete cost-ordered deletion sweeps. For nondegenerate references, each accepted deletion maintains feasibility, and a nonempty terminating subset admits no individually removable retained attribute under the specified constraint. A label-margin condition further provides a sufficient guarantee for preserving a unique dominant label.

Experiments on 25 tasks spanning 16 feature families, together with a separate image-pipeline study, demonstrate individual training-reconstruction deterioration under mean preservation and characterize the acquisition cost of enforcing objectwise bounds. Matched-constraint comparisons show that search strategy can affect retained acquisition cost, while geometry, training-sample size and tolerance alter the reduction outcomes. The specified deletion search can retain the full set under strict preservation; relaxed bounds quantify storage–reconstruction trade-offs within the evaluated candidate pools. These findings support evaluating explicitly defined individual reconstruction requirements while assessing held-out prediction and measured resource use separately.

Notation and abbreviations

The tables collect the principal symbols and method labels. Reconstruction allowance ϵ_{rec} is distinct from split-radius threshold η_R, and scientific tolerance τ differs from attribute bandwidth τ_a.

Table 21: Principal mathematical notation

Symbol	Meaning
O, o_i, n	Training objects, the ith object and the number of objects.
A, m, P, Q	Full attributes, $m = \lvert A\rvert$, current subset and candidate subset. The denominator m stays fixed.
A_n, A_c	Numerical and categorical attributes.
$\mathcal{Y}, q$	Labels and their number.
$L, \ell_i^k, \boldsymbol{\ell}_i$	Label-distribution matrix, description degree and object label vector.
$c_a, C(P) = c(P)$	Positive simulated attribute price and additive subset cost.
$\boldsymbol{p} = \bar{\boldsymbol{\ell}}$	Global mean training distribution and fallback reconstruction.
z_a, K_a, τ_a, β	Scaled value, affinity, kernel bandwidth and shrinkage parameter.
$\boldsymbol{u}_a, \widetilde{\boldsymbol{u}}_a$	Raw and regularized local label profiles.
W_a, B_L	Local KL-divergence weight and dataset normalizer.
Δ_a, G_a, ζ	Hybrid feature discrepancy, raw geometric discrepancy and mixture coefficient.
$\overrightarrow{\rho}_P, \overleftarrow{\rho}_P, d_P$	Forward, backward and symmetrized dissimilarities.
$B, \boldsymbol{\mu}_B, V(B)$	Granular ball, mean label vector and within-ball dispersion.
$\mathrm{Cons}(B)$	Consistency $1 - V(B)$; distinct from acquisition cost $C(P)$.
$V_0, s_{\min}, \kappa, \alpha$	Global dispersion, minimum support, relative dispersion tolerance and radius bound.

Table 21 (continued)

Symbol	Meaning
η_V, η_R	Nonnegative split-gain thresholds for dispersion and radius.
$\mathrm{Eligible}_P(B)$	Node eligibility predicate, independent of the final leaf set.
$\mathcal{E}_P^0, \widetilde{\mathcal{E}}_P$	Base eligible anchors and final accepted anchored leaves.
$\Gamma(P), \widetilde{\Gamma}(P)$	Base and anchored aggregate reconstruction dependencies.
$K, \epsilon_{\mathrm{safe}}$	Seed-pair budget per node, including the first pair, and new-leaf safety slack.
$\widetilde{\boldsymbol{\ell}}_i(P), e_i(P)$	Anchored reconstruction and squared error against the observed target.
$e_i^\star = e_i(A)$	Fixed full-attribute reconstruction error.
$\tau, \epsilon_{\mathrm{num}}, \epsilon_{\mathrm{rec}}$	Scientific tolerance, numerical slack and allowance $\tau + \epsilon_{\mathrm{num}}$.
$\mathcal{F}_{\mathrm{O1}}, P^\star$	O1-feasible family and global cost minimizer; the deletion algorithm does not certify that minimizer.
$J, \mathrm{changed}$	Ordered attributes at sweep start and flag for an accepted deletion.
γ_i	Gap between the largest and second-largest target degrees, for a unique maximum and $q \geq 2$.
$B(P)$	Held-out Brier error; distinct from a granular ball B.
$\mathcal{C}, d(P), S_{\mathcal{C}}(\tau)$	Recorded candidate pool, largest nonnegative error increase and minimum recorded feasible storage.

Table 22: Abbreviations and method labels

Term	Meaning or usage
LDL	Label distribution learning.
O1	Fixed-reference objectwise reconstruction-error constraint.
GBIP-O1	Granular-ball information-preserving reduction under O1; complete-sweep cost-ordered deletion with the stated reconstruction-error meaning of preservation.
GBFRS	Granular-ball fuzzy rough-set selector of Qian et al. [9].
GBFRS-EQ	This study's equation-based GBFRS implementation under the disclosed protocol.
GBFRS+O1	Least-cost feasible prefix of the GBFRS-EQ ranking under the shared evaluator.
IWDLFS, IWCLFS	Dependence-based and conditional-discrimination-based informational weighted selectors [10].
Mean	Complete-sweep comparator using average training-error preservation.
Full, Empty	All attributes or no attributes; Empty predicts the training-label mean.
AA-kNN	Algorithm-adaptation k-nearest-neighbor LDL predictor.
CV, PCA	Cross-validation and principal component analysis.
KL, TV, JS	Kullback–Leibler divergence, total variation and Jensen–Shannon divergence.
MAE	Mean absolute error.
CPU, MB	Central processing unit and decimal megabyte (10^6 bytes).

Declarations

Originality Statement

This manuscript has not been published previously and is not under consideration for publication elsewhere. All content is original; no substantial parts have been published or submitted to another journal, and all third-party material is properly cited.

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Competing Interests

The authors declare no financial or non-financial competing interests relevant to this work.

Ethics Approval

This work is a secondary computational analysis of publicly available LDL benchmark datasets (e.g., Natural Scene, SBU-3DFE, the Yeast gene-expression series, SJAFFE, FG-Net, Movie, and Human Gene; the complete list with sources is given in Table 1). No experiments involving human participants or animals were performed by the authors, and no new participant data were collected. Information on the original data collection, ethics review and permitted uses should be consulted in the original dataset publications and the terms supplied by the respective data custodians.

Consent to Participate and Consent for Publication

Not applicable. This study uses only publicly available benchmark datasets; no new participant data were collected by the authors. The original data collection and release are governed by the consent procedures and terms of the respective data custodians; availability of a dataset does not itself establish unrestricted redistribution rights.

Data Availability

The label distribution learning benchmarks analysed in this study are obtained from the LDL-LDM and PyLDL repositories and, for the separate SCUT-FBP5500 image-pipeline study, the download links provided by the official SCUT-FBP5500 repository. Source URLs, repository commits, and file hashes are recorded in the dataset source manifests within the reproducibility archive. Reanalysis scripts and aggregate results supporting the findings are provided in the public reproducibility archive. Third-party raw datasets are obtained from their upstream sources under the applicable access and use terms; they are not redistributed in the public code archive.

Code Availability

The source code and frozen result files supporting the conclusions of this article are archived at `https://github.com/barbarossa-li/ldl-cost-sensitive-reduction`. The repository is publicly accessible; versioned protocols, source manifests, checksums and aggregate results identify the corresponding experimental records.

Author Contributions

Chaoyang Li: Investigation, scoring execution, formal analysis, writing—original draft preparation. **Xiaohan Li**: Conceptualization, methodology, visualization, validation, resources. **Zhaowen Li**: Data curation, supervision, project administration, writing—review and editing, funding acquisition. All authors have read and approved the final version of the manuscript.

Use of Generative AI

Generative AI tools (OpenAI ChatGPT/Codex) were used to assist with language editing, manuscript revision, literature and protocol discussions, code development and debugging, and analysis of computational outputs. The reported numerical results are derived from the recorded computational runs and their saved outputs. AI-generated suggestions and text require author verification against the source literature, code and experimental records. The authors retain responsibility for the accuracy, originality and integrity of the manuscript, including all methods, results, citations and conclusions.

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